

Cyclic Slip Irreversibilities and the Evolution of Fatigue Damage

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In this article, the physical origin of fatigue crack initiation in ductile metals is discussed from a historical perspective. The main focus is to assess those cyclic slip irreversibilities in a microstructural sense that occur not only at the surface but also in the bulk at the dislocation scale and to show how they contribute to surface fatigue damage. The evolution of early fatigue damage, as evidenced experimentally in the last decades, is reviewed. The phenomenon of cyclic strain localization in persistent slip bands (PSBs) and models of the formation of extrusions, intrusions, and microcracks are discussed in detail. The predictions of these models are compared with experimental evidence obtained on mono- and polycrystalline face-centered-cubic (fcc) metals. In addition, examples of the evolution of fatigue damage in selected fcc solid solution alloys and precipitation-hardened alloys and in body-centered-cubic (bcc) metals are analyzed. Where possible, the cyclic slip irreversibilities p , defined as the fraction of plastic shear strain that is microstructurally irreversible, have been estimated quantitatively. Broadly speaking, p has been found to vary over orders of magnitude ($0 < p < 1$), being almost negligible at low loading amplitudes (high fatigue lives) and substantial at larger loading amplitudes (low fatigue lives).

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Hael Mughrabi obtained his primary school education in Germany and subsequently attended different British schools in Jerusalem, Palestine, and in Cairo, Egypt, where he graduated from secondary school at The English School in 1955. He is a German citizen. After a three-year apprenticeship as a mechanic and toolmaker at Bosch GmbH Company in Stuttgart, he entered the Engineering University of Stuttgart to study physics. During his studies, in his doctoral thesis work and in his subsequent occupation as a senior researcher at the Max-Planck-Institute of Metal Research in Stuttgart (1966–1983), he specialized in metal physics and performed mainly research in the fields of crystal defects, mechanical properties, and microstructural characterization. He held a Visiting Professorship at Cornell University in 1978–1979. After receiving offers from three universities, he joined the University of Erlangen–Nürnberg in 1984 as a Professor of Materials Science and Engineering and Head of an Institute of General Materials Properties. In subsequent years, he held positions as Department Head

and Dean of the School of Engineering. Since 2002, Hael Mughrabi has been formally retired but is still active in various forms in research and in committee work.

Hael Mughrabi has published almost 300 papers and book chapters and has been editor or co-editor of several books and conference proceedings, mainly in the fields of crystal plasticity, materials characterization, metal fatigue, high-temperature mechanical properties of nickel-base superalloys, and modeling of mechanical behavior. He has been active in the organization and chairing of many international conferences and has frequently been invited as a plenary or keynote speaker. Hael Mughrabi has been a member of TMS for 25 years and of a number of other professional societies, including the German Materials Society (Deutsche Gesellschaft für Materialkunde (DGM)). He has been a member of the Board of Directors of DGM and Chairman of the DGM Awards Committee PKII and has been elected an Honorary Member of DGM. Hael Mughrabi is the holder of an Honorary Doctoral degree from the Ruhr University Bochum and the recipient of several national and international awards, including the highest award of DGM, the Heyn-Denkünze.

I. INTRODUCTION

THE study of fatigue failure of metals has been one of the most important topics of interest to materials scientists for approximately the last 150 years. In the middle of the 19th century, the more frequent occurrence of failures caused by repeated cyclic loadings motivated the first systematic studies of the hitherto unknown phenomenon of metal fatigue. The Frenchman, Jean Victor Poncelet, apparently was the first to describe this behavior by saying that the material had become tired and he coined the "term fatigue." The early studies of fatigue failures are closely linked with names such as Wilhelm Albert,^[1] Sir William Fairbairn,^[2] and August Wöhler.^[3] The portraits of the latter two are shown in Figure 1. Wöhler, in particular, became famous for having conducted the first systematic studies of metal fatigue. It is less well known that Johann Bauschinger^[4] (also shown in Figure 1), discoverer of the well-known effect that the yield stress is lowered after stress reversal, also performed some detailed studies of metal fatigue in which he substantiated Wöhler's observations and formulated some general conclusions (compare Reference 5). The main aim of these early studies was to establish suitable laboratory testing techniques and to determine after how many repeated loadings of a given type failure occurred and to thus characterize fatigue life in terms of the numbers of cycles to failure, N_f . In these early studies, little attention was paid to the microscopic origin of fatigue failure. The first insight into the mechanisms of metal fatigue was obtained by Sir James Alfred Ewing, also shown in

Figure 1, and Humfrey in 1903.^[6] These authors demonstrated that slip bands, which gradually developed into cracks, appeared on the surface of steel specimens that had been subjected to repeated cyclic loadings at stresses that were considered so small as to lie below the elastic limit. It thus became gradually evident that fatigue damage in the form of irreversible microscopic changes at the surface resulted from the accumulation of very small irreversible plastic cyclic microstrains and that metal fatigue was actually a problem of microplasticity. Nonetheless, it was not until 1953, when Coffin^[7] and Manson^[8] showed independently that fatigue life depended on the plastic strain amplitude, that plastic strain was considered important. Today, it is clear that mechanisms of cyclic microplasticity, based on the glide of dislocations, are responsible for the fatigue phenomena.

Since the discovery of the microscopic nature of the evolution of fatigue damage by Ewing and Humfrey more than 100 years ago and, 50 years later, the recognition of the importance of plastic strain by Coffin and Manson, the microstructural mechanisms leading to fatigue damage have been investigated thoroughly. Thus, today, a much more detailed understanding of the microstructural mechanisms leading to fatigue damage and failure of metals has been achieved. These advances have been made possible by the tremendous progress achieved in the instrumentation of fatigue testing, in the techniques of microscopic analysis of fatigue damage, and in the computational facilities. Details on these developments can be found in several monographs on metal fatigue^[9-12] and in a number of reviews that appeared in the last decades.^[13-23] In spite of these developments, open questions remain and continue to motivate more and more detailed research directed at the elucidation of the microscopic processes that lead to fatigue damage and, ultimately, to fracture and failure. Thirty years ago, Professor Morris Fine, from Northwestern University, held the Edward DeMille Campbell Memorial Lecture and selected "Fatigue Resistance of Metals" as the title of his talk.^[24] In his lecture, Professor Fine addressed the following subjects:

- (1) property and microstructural changes due to cyclic plastic deformation,
- (2) initiation of microcracks,
- (3) growth of microcracks to form a macrocrack, and
- (4) propagation of macrocracks from threshold to near final failure.

The present work will only deal with the first two topics and will be confined to fatigue phenomena in ductile metals in which fatigue damage usually occurs at the surface and then spreads into the bulk. An attempt will be made to provide a historical perspective of the developments over the last decades and to outline our present understanding, as seen by the author. One main goal will be to identify for different materials the microscopic mechanisms of cyclic deformation that lead to the *microstructurally* irreversible evolution of those features at the specimen surface from which cracks originate.

Sir William Fairbairn



August Wöhler



Johann Bauschinger



Sir James Alfred Ewing

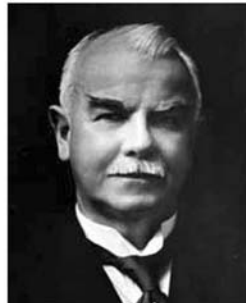


Fig. 1—Portraits of Sir William Fairbairn, August Wöhler, Johann Bauschinger, and Sir James Alfred Ewing.

II. ORIGIN AND DEFINITION OF CYCLIC SLIP IRREVERSIBILITIES

Since fatigue damage generally initiates at the surface, it is natural to first think of cyclic slip irreversibilities at the surface. Then, any dislocation that emerges at the surface during forward loading and that does not return into the material during reverse loading will leave an unreversed slip step behind. The accumulation of unreversed slip steps on the surface can lead to a gradual roughening of the surface, with sites of local stress concentrations at which microcracks can form. While these features of cyclic slip irreversibilities at the surface appear straightforward, it is more difficult to understand that there are also slip irreversibilities that occur in the bulk but, nonetheless, convey the information, frequently in a less direct manner, to the surface and thus contribute to the evolution of fatigue damage, as will be discussed in more detail later. More specifically, the following *microstructural* mechanisms will be considered:

- (1) cross-slip of screw dislocations, implying that forward and reverse glide of dislocations do not occur on the same slip planes;
- (2) mutual annihilation of unlike screw and unlike edge dislocations: whenever a dislocation is annihilated, this marks the end of its glide path, implying that reverse deformation cannot occur along the same glide path;
- (3) so-called slip plane asymmetry in the case of body-centered-cubic (bcc) metals: in this case, forward and reverse glide of dislocations occur on different glide planes in tension and in compression; and
- (4) random to-and-fro glide of dislocations.

All these slip irreversibilities will manifest themselves in one form or another at the surface and will contribute to the gradual evolution of fatigue damage. It is expedient to characterize the degree of cyclic slip irreversibility by some quantity. Here, the cyclic slip irreversibility p will be defined, as in earlier work,^[25–35] as the fraction of plastic shear deformation that is irreversible in a *microstructural* sense:

$$p = \frac{\text{irreversible plastic shear strain}}{\text{cumulative total plastic shear strain}} \quad [1]$$

The thus defined cyclic slip irreversibility p will be zero when slip is entirely reversible, and it will be equal to one when slip is entirely irreversible. Hence, $0 < p < 1$. Moreover, the cyclic slip reversibility p will depend on the plastic shear strain amplitude $\gamma_{pl} = \Delta\gamma_{pl}/2$ ($\Delta\gamma_{pl}$: plastic shear strain range), *i.e.*, $p = p(\gamma_{pl})$, p being the larger, the larger the shear strain amplitude γ_{pl} is. The reason is simply that small to-and-fro dislocation displacements will be more reversible in the preceding sense than larger ones. The irreversible part of the plastic shear strain amplitude that is responsible for “unreversed” microstructural changes can be written as

$$\gamma_{pl,irr} = p \times \gamma_{pl} \quad [2]$$

Then, a *cumulative* irreversible plastic shear strain after N cycles can be defined as^[30]

$$\gamma_{pl,irr,cum} = 4N \times p \times \gamma_{pl} \quad [3]$$

In order to be able to discuss different examples of slip irreversibility under similar assumptions, this definition will be used in the following. In the literature, other forms of definition of cyclic slip irreversibility can be found, as detailed, for example, in the works of other authors.^[36–42] It should be noted that these latter definitions are all based on the irreversibility of slip step formation *at the surface*. The advantage of this definition is, of course, that it refers directly to the site of crack initiation. In contrast, the definition given by Eq. [1] is based on the assessment of cyclic slip irreversibilities *in the bulk*, which, at first sight, are less directly related to the fatigue crack initiation process. However, as will be shown subsequently, this bulk cyclic slip irreversibility can in fact be related to the phenomena at the surface, if the dislocation mechanisms in the bulk are taken into account appropriately. In general, it is not an easy task to assess quantitatively the cyclic slip irreversibility. For this reason, it has so far only been possible to obtain quantitative estimates of the slip irreversibility p in a limited number of cases, which had been characterized in sufficient detail experimentally. In the following, some of these cases will be described in more detail.

III. MATERIALS CONSIDERED

The examples to be discussed will refer to fatigue of some selected model and engineering materials, which have been shown to be suitable for basic studies.

- (1) pure face-centered-cubic (fcc) metals such as copper or nickel, with so-called “wavy slip” character^[15,17,43] in which cross-slip is easy, leading to almost complete annihilation of screw dislocations and clustering of dipolar edge dislocations at low to intermediate loading amplitudes and dislocation cell formation at higher loading amplitudes;
- (2) solid-solution-hardened fcc alloys, such as α -brass, with so-called “planar slip” character^[15,17,43] in which cross-slip is strongly impeded, leading to the formation of planar dislocation groups/arrays on one slip system at low and on more slip systems at higher loading amplitudes (the more general factors that govern wavy/planar slip have been discussed in References 44 and 45);
- (3) precipitation-hardened fcc alloys with wavy slip, containing very small shearable particles, or with planar slip, containing larger shearable particles (compare References 21 and 27);
- (4) pure bcc metals, such as α -iron or niobium; and
- (5) solid-solution-hardened bcc alloys, including some simple ferritic steels.

IV. EARLY WORK ON CYCLIC STRAIN LOCALIZATION AND FORMATION OF EXTRUSIONS AND INTRUSIONS

A. Formation of Persistent Slip Bands

The occurrence of fatigue-induced slip bands was noted for the first time in 1903 by Ewing and Humfrey in their pioneering microscopic study of Swedish iron that had been fatigued in rotating bending.^[6] Subsequently, many others studied fatigue-induced slip bands in more detail, in particular, Gough and co-workers,^[46] who showed, among other things, that the slip bands occurred on the same crystallographic planes on which deformation occurred in unidirectional straining. Thompson *et al.*,^[47] more than 50 years after the work of Ewing and Humfrey, studied the nature and properties of slip bands in fatigued copper mono- and polycrystals in impressive detail and showed that the slip bands were “persistent” in the sense that they reappeared at the old sites, when the specimen was fatigued again after the previously formed slip bands had been polished away. Thus, they could show that these “persistent slip bands” (PSBs), as they called them, were not merely a surface but a bulk phenomenon. This was confirmed convincingly later by Laufer and Roberts^[48] in their study of fatigued copper single crystals by transmission electron microscopy (TEM). These authors showed that the dislocation arrangement in PSBs in the bulk was distinct from that of the surrounding so-called matrix and consisted of the so-called wall or ladder structure, which was embedded in the matrix structure of dipolar dislocation bundles or veins, as will be discussed in more detail later.

It is clear from the work of Brown's group in Cambridge,^[49–51] Finney and Laird,^[52] and the author,^[20,53,54] on copper monocrystals of single-slip orientation, fatigued in strain control, that PSBs appear in the approach to cyclic saturation. Moreover, it has been shown that well-defined threshold amplitudes exist, below which PSBs are not usually formed.^[53,55,56] As the cyclic strain becomes localized in the PSBs, a complex rough surface relief develops at the traces of emerging PSBs, whereas the surface between the PSBs remains almost featureless. Fatigue cracks then initiate at sites of stress concentration in the surface relief of emerging PSBs. In general, this sequence of events has been regarded as typical of and as a necessary prerequisite for fatigue crack initiation in ductile fcc metals.^[13–15,17–23,47,57] On the other hand, different views have been advocated on the detailed nature of the surface relief and the mechanisms by which it develops, and these issues are still under discussion today,^[19,23,26,58–61] as will be discussed later. The threshold amplitudes for the formation of PSBs have been related to the existence of a true fatigue limit.^[58,62] More recently, however, it has become clear that, in all probability, these thresholds represent only the condition for the initiation of stage I shear microcracks but do not necessarily fulfill the conditions for the subsequent propagation of these cracks, as emphasized by several authors.^[33–35,63,64] Hence, according to these studies, the

true fatigue limit actually lies somewhat above the thresholds for PSB formation.

B. Cyclic Strain Localization and Irreversibility of Cyclic Slip in PSBs within Cycles

The question whether cyclic strain localization in PSBs remains constant during cycling and how reversible/irreversible cyclic slip is within one or more cycles has been investigated by several authors who used different techniques. Again, these studies were mainly performed on copper crystals^[52,65–68] but more recently also on nickel single crystals^[69,70] and polycrystals.^[71,72] Thus, Watt *et al.*^[65] and Basinski *et al.*,^[66] using optical microscopy, concluded that slip steps generated in one half-cycle almost disappeared in the subsequent reverse half-cycle. In subsequent more precise measurements, Finney *et al.*,^[68] who used surface interferometry, showed that reversal of slip steps was only partial and that the strain localization differed strongly within a PSB and was, in fact, larger by an order of magnitude in some regions. These results were substantiated by the Basinskis^[22,67] with respect to the nonuniformity of cyclic strains within PSBs and the much larger *local* strains in so-called persistent slip “lines.” Even more detailed information was obtained by the Dresden group^[69,70] by atomic force microscopy on fatigued nickel single crystals, who quantified an entire spectrum of *local* strains within PSBs and emphasized that considerable variations in local strain occurred within one and the same PSB as cycling was continued. Subsequently, Weidner *et al.* combined atomic force microscopy (AFM) and SEM and characterized the half-cycle slip activity in PSBs in nickel polycrystals and the variations in the course of fatigue in considerable detail.^[71,72] From these studies, it follows that cyclic slip in PSBs is a highly complex process. This should be kept in mind in the following discussion of the dislocation mechanisms in PSBs, the evolution of the surface profile, and the development of models that are, of necessity, not able to capture all the details that have been revealed in experiments. Nonetheless, it is hoped that the models reflect the governing main features in a satisfactory manner.

C. Surface Relief Observed after Fatigue

Early studies of the surface relief were mainly confined to optical microscopy and later to scanning electron microscopy (SEM) observations of the surface or of sections under an angle to the surface. Perhaps the first spectacular evidence of dramatic changes in the surface relief of PSBs in fatigued metals was Forsyth's observation of thin sheets of metal being extruded some microns out of PSBs in a fatigued age-hardened aluminum alloy.^[73] Aside from that, Wood's studies of so-called taper sections,^[57,74] performed on different fatigued metals, deserve special mention. By sectioning the fatigued specimen under a rather small acute angle to the surface and perpendicular to the traces of the slip bands, an appreciably magnified picture of the vertical displacements at the surface could be obtained, as

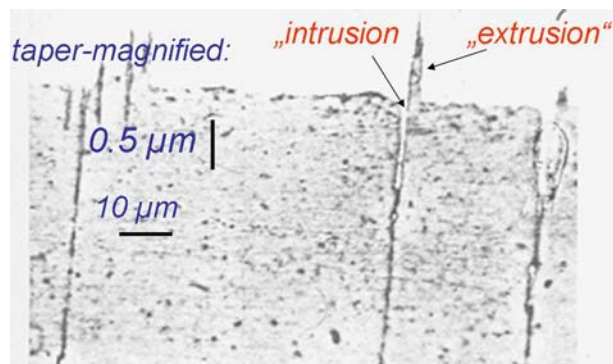


Fig. 2—Example of optical micrograph of taper section of fatigued copper polycrystal, showing “notches and peaks” forming at the head of a slip band.^[57]

shown in the example in Figure 2. Wood’s observations that cracks initiate in the notch-peak surface geometry formed by to-and-fro displacements led to May’s model of surface roughening by random slip.^[75] While the assumption of random slip may be criticized, it is a useful concept to formulate, in an approximate manner, the evolution of surface roughness and has also been made use of in this work. The studies by Hull^[76] and Cottrell and Hull,^[77] who used the high resolution of the electron microscope replica technique to image the surface relief of fatigue copper mono- and polycrystals, were equally important. In particular, these studies showed that the surface relief developed also at low temperatures (4.2 K), indicating that the evolution of the surface relief occurred by a purely mechanical mechanism. In these early observations, it was found that so-called extrusions (elevations) developed along the PSBs at the surface and that there were also intrusions (depressions). Moreover, in some of the observations (compare Figure 2), the impression was gained that extrusions and intrusions developed side by side, *i.e.*, in the form of extrusion-intrusion “pairs,”^[14,74,76,77] suggesting that intrusions and extrusions form simultaneously and that the intrusions formed somehow by an “inverse” mechanism of extrusion formation.

Several attempts were made to interpret these observations. Within a few years, a number of dislocation models were put forward,^[77–82] most of which were designed to predict the simultaneous formation of extrusions and intrusions in the form of extrusion-intrusion pairs. These models have been discussed in detail in subsequent reviews.^[9,14,82] Here, we shall only make some general comments. Further, to give a flavor of these models, all of which were based on rather special assumptions, the early models of Cottrell and Hull^[77] and Mott,^[78] which were devised before the first TEM observations of the dislocation distributions had been made, will be discussed. Then, the later model of Watt,^[81] in which use was made of the knowledge of the dislocation arrangements observed by TEM observations, will be presented. These three models are sketched in Figures 3 through 5. In the model of Cottrell and Hull (Figure 3), two dislocation sources on different

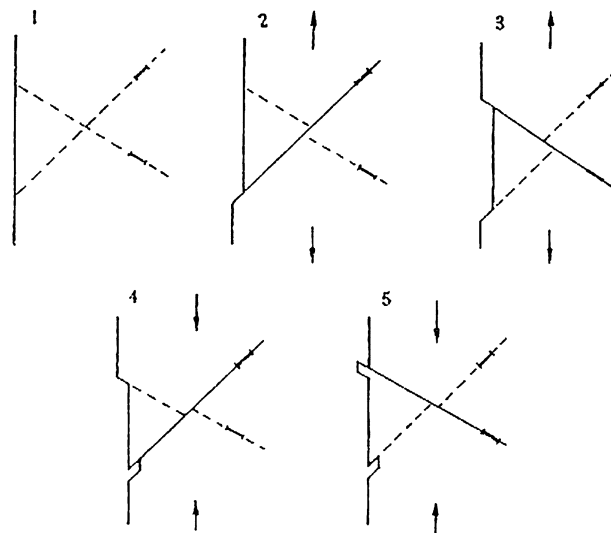


Fig. 3—Sequence of slip movements producing an extrusion and an intrusion.^[77]

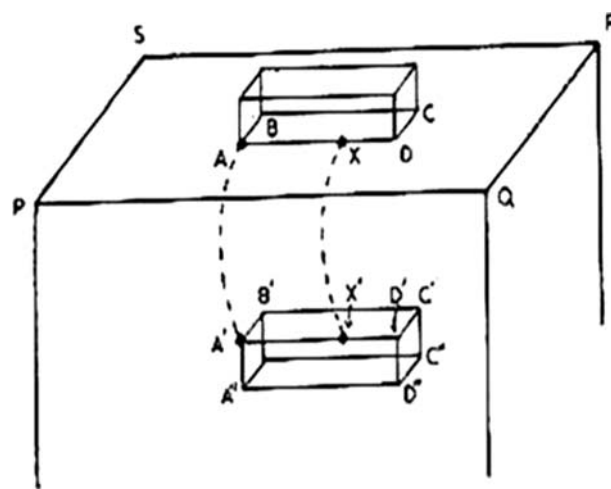


Fig. 4—Mott’s model of extrusion formation.^[78]

glide planes operate in sequence, first in the forward and then in the reverse half-cycle. Thus, an extrusion and an intrusion are created. Shortcomings of the model are not only that the extrusion and the intrusion do not appear on the same slip system and that they are isolated and spaced apart, but also that the operation of the model depends on very special assumptions. These criticisms apply also to the models of McEvily and Machlin^[79] and Thompson,^[80] which will not be discussed in more detail. Mott’s model (Figure 4) was the first of several models in which the cross-slip of screw dislocations played an important role. This model is essentially only a model of extrusion formation. The basic idea is that a screw dislocation, which has one end terminating at the free surface while the other end terminates at a voidlike cavity in the interior, slips and cross-slips around a trapezoid in a closed path, whereby an extrusion is formed by the slip steps generated at the surface, while at the same time, the internal void increases in size.

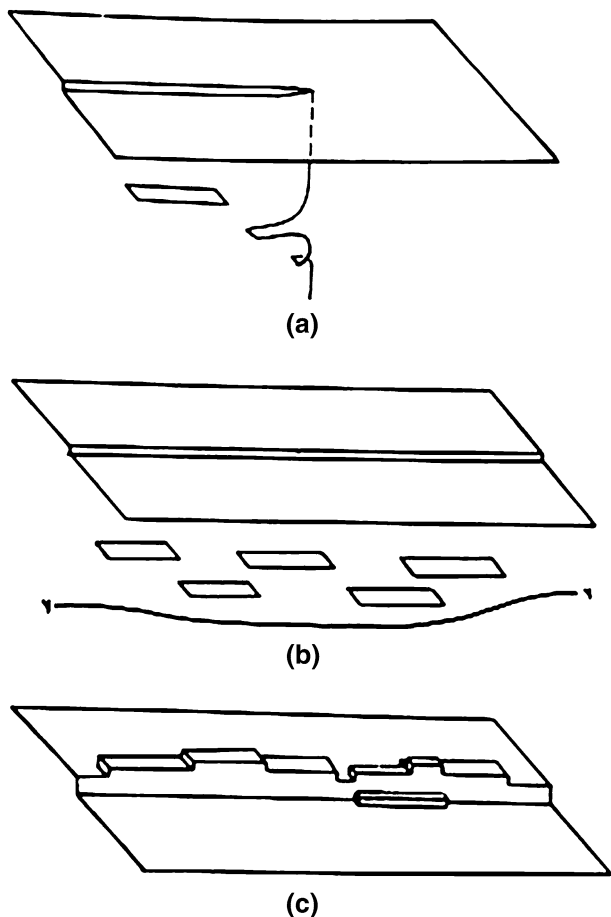


Fig. 5—Watt's model of the evolution of the surface relief: (a) dipole formation, (b) dislocation approaching and driving the dipoles to the surface, and (c) emergence of dipoles and creation of surface relief.^[81]

Mott, in fact, assumed that the internal void originated in the dilation field of two very close edge dislocations in such a way that a crack would be formed. Thus, Mott's basic idea of extrusion formation is that matter is added to the surface by a dislocation mechanism, while in the interior, voidlike cavities of equivalent volume develop. This view, slightly modified, has also been favored by Boettner and McEvily, who found mainly extrusions (and almost no intrusions) in fatigued silicon iron.^[83] Except for the details concerning the nature of the voids and the actual crack initiation mechanism, Mott's basic idea of extrusion formation can still be considered as realistic today. As pointed out by Kennedy,^[82] it is not readily seen why the dislocation in Mott's model does not simply glide backward and forward instead of traveling by cross-slipping around a closed circuit. Furthermore, the model cannot be extended easily to also explain the formation of intrusions. For this reason, Kennedy has added Lomer–Cottrell dislocations to the picture and introduced what he calls a “gating mechanism.” Then, with a number of assumptions, it is argued that the dislocation will be forced to travel on a closed path and that both ex- and intrusions can be formed. One can summarize the essence of these models by

saying that they contain elements, which are plausible and could function in practice, whereas, at the same time, rather specific assumptions are made, which make it unlikely that the models would function in reality. In other words, there must be some more general mechanisms that would function without requiring rather specific assumptions.

Finally, attention is now turned to the model of Watt,^[81] which, being based on TEM observations, differs in an important way from the earlier models. Those observations showed that, in copper, fatigued at low to intermediate loading amplitudes, the dislocation arrangements formed initially during cyclic hardening consisted of dislocation clusters, *i.e.*, so-called “veins” made up essentially of glissile elongated edge dislocation dipole loops of either vacancy or interstitial character, as discussed in more detail subsequently. When such elongated loops are swept to the surface, an interstitial- or vacancy-type dipole will leave behind an embryonic extrusion or intrusion, respectively. It is then envisaged that, when these processes occur in a statistical manner, a surface relief consisting of extrusions and intrusions will develop during continued cyclic deformation. Models of this type, which take into account the actual prevailing dislocation distribution, appear more realistic than the other models described previously. However, in order to develop a more comprehensive model of the evolution of the surface relief, the dislocation glide processes in PSBs must first be analyzed in more detail.

V. DETAILED ANALYSIS OF DISLOCATION GLIDE PROCESSES IN PSBs IN FCC METALS

A. Glide and Annihilation Mechanisms

The formation of the dipolar edge dislocation clusters, *i.e.*, the veins, during cyclic deformation of fcc metals, is easily visualized, when one considers that screw dislocations of opposite sign can easily annihilate mutually by cross-slip. The remaining edge dislocations assemble in clusters of elongated dipolar loops, which can be of vacancy or interstitial type and which are frequently chopped by cross-slip into loop segments of shorter length, terminated by the superjogs left behind on the cross-slip plane. The elongated dipole loops cluster together to form the di-/multipolar veins, as described in more detail earlier.^[16] Thus, cross-slip is important in two ways: elimination of the screw dislocations and subdivision of the elongated dipole loops into shorter segments. After cyclic hardening has progressed to the stage at which the work-hardened dislocation distribution can no longer accommodate the imposed plastic strain, the vein structure breaks down locally, giving way to the softer newly formed lamellae of the PSB wall structure, lying parallel to the primary glide plane. The surface appearance of PSBs in a fatigued copper polycrystal is shown in Figure 6(a). The surface had initially been electropolished. After fatigue, the PSBs are clearly visible by SEM as a set of parallel bands with a roughened surface, running along the trace of the active primary slip system. In contrast to the rough surface of

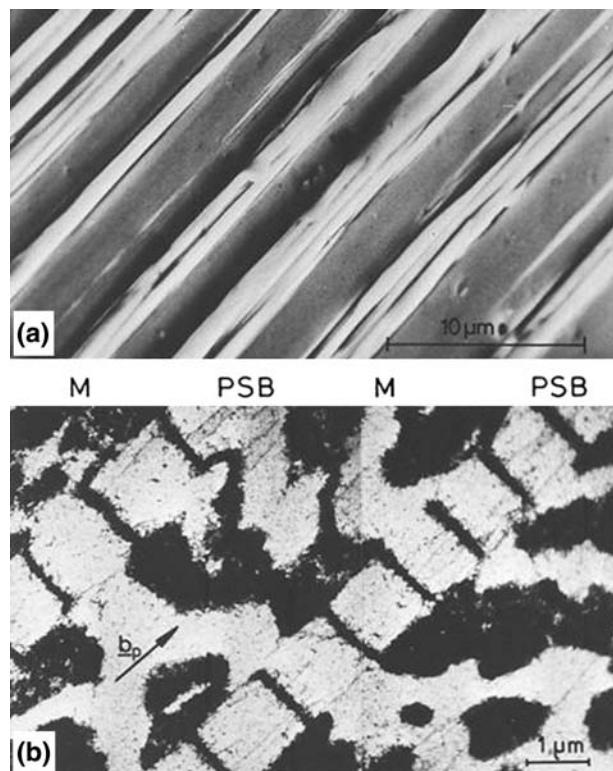


Fig. 6—PSBs in fatigued copper: (a) SEM, surface relief; (b) TEM micrograph of PSBs with ladder structure between matrix of dipolar veins in a section perpendicular to the primary slip plane, containing primary Burgers vector b_p .^[21]

the PSBs, the surface between the PSBs is featureless, indicating that, after formation of the PSBs, the cyclic strain is essentially localized in the PSBs. Details of the dislocation arrangements in the PSB lamellae and in the remaining “matrix” of veins are shown in Figures 6(b) and 7. In a section perpendicular to the glide plane of the operating primary glide system and to the line direction of the primary edge dislocations (*i.e.*, the primary Burgers vector b lies in the plane of the figure), the dislocation arrangement in the bulk of a fatigued copper single crystal is as shown in Figure 6(b). Figure 7 shows, in a three-dimensional (3-D) TEM montage, the spatial arrangement of the dislocation distribution.^[56] In both figures, the ladderlike PSBs, separated by the veins of the matrix structure, can be recognized clearly. The vein and the wall structures have been studied in considerable detail in fatigued copper single crystals by several groups.^[14,17,51–56,84] Broadly speaking, the dislocation distribution in fatigued nickel single crystals, studied extensively by the group in Dresden,^[85,86] is very similar. In the veins and in the walls, the local dislocation density is very high (some 10^{15} m^{-2}). Hence, individual dislocations are difficult to resolve with standard TEM magnification, and both veins and walls appear as black clusters in Figures 6 and 7.

The mechanisms of dislocation glide in the PSB wall structure are understood best by viewing the dislocation distribution in the wall structure in a section parallel to the primary glide plane, as shown in Figure 8. One can

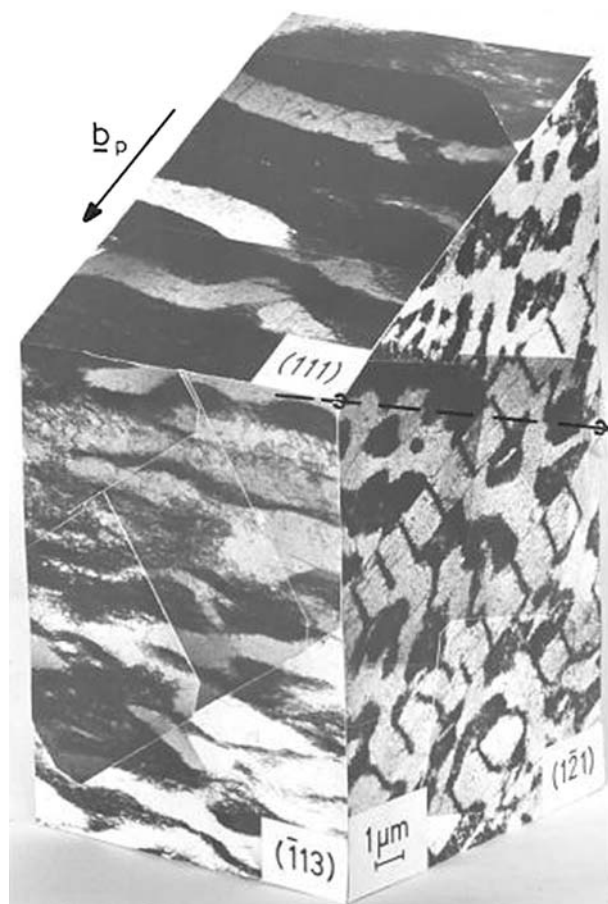


Fig. 7—3-D montage of TEM micrographs, showing PSBs with ladder structure and dipolar matrix veins in fatigued copper single crystal.^[56]

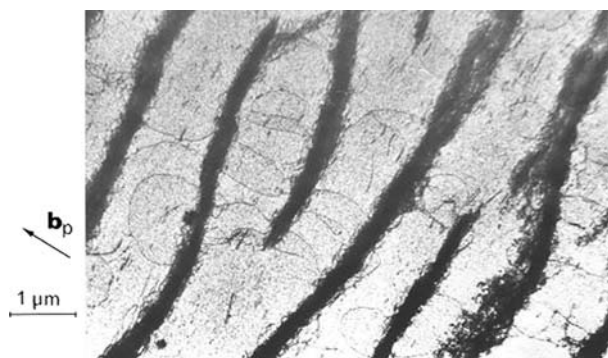


Fig. 8—TEM micrograph of section parallel to primary glide plane (111) of fatigued copper single crystal, showing edge dislocation walls of PSBs, with small edge segments bowing out of the walls and bowed-out screw dislocations gliding in dislocation-poor channels.^[21,56]

see bowed-out screw dislocations gliding along the channels between the walls and also bowed-out edge segments curving out of the walls and traversing the channels. The bowing-out edge segments lead to an increase of the dislocation density by producing new screw segments across the channels, which then glide

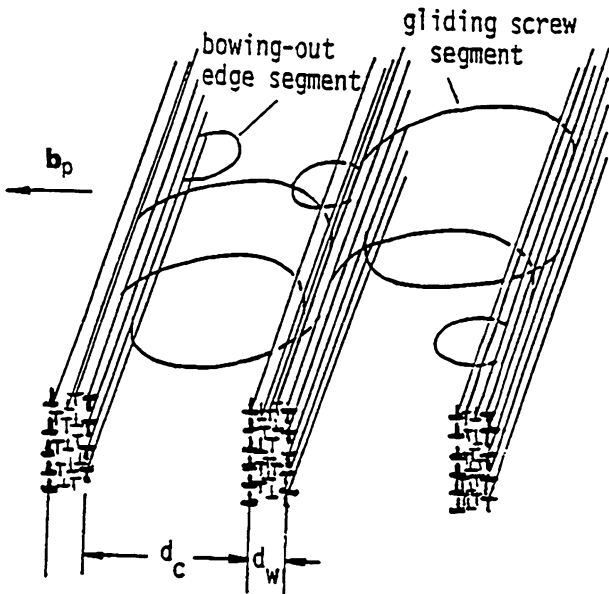


Fig. 9—Schematic illustration of dislocation glide processes in the wall structure of PSBs in fcc metals.^[20,21]

along the channels and deposit new edge dislocations in the walls, as illustrated schematically in Figure 9.

Since the dislocation patterns discussed correspond to a state of cyclic saturation, it follows that the increase of the dislocation density *must* be balanced precisely by corresponding mutual annihilation processes of both screw and edge dislocations, as described in detail previously.^[54,56,87,88] It must be recalled that Backofen had already come to the same conclusion in 1959,^[89] although the precise dislocation distribution was not known at that time and had also concluded, that, in cyclic saturation, both screw and edge dislocations must annihilate. As detailed elsewhere, experimental evidence exists showing that, in copper fatigued at room temperature, two screw dislocations of opposite sign that encounter each other on glide planes closer to each other than the so-called annihilation distance $y_s \approx 50$ nm will annihilate spontaneously by cross-slip.^[56] In a similar fashion, closely spaced edge dislocations of opposite sign will annihilate mutually in an athermal manner, when the glide planes are spaced closer than the corresponding edge annihilation distance $y_e \approx 16$ nm.^[87,90] The detailed mechanism by which this occurs is admittedly not yet well understood.

B. Model of Reversible and Irreversible Dislocation Glide in PSBs

In a simplified picture, the annihilation processes of screw dislocations can be considered as the main origin of slip irreversibility in the PSBs.^[25,56] The shear strain amplitude is given by

$$\gamma_{pl} = \pm \rho \cdot b \cdot L \quad [4]$$

i.e.,

$$\rho = \gamma_{pl} / b \cdot L \quad [5]$$

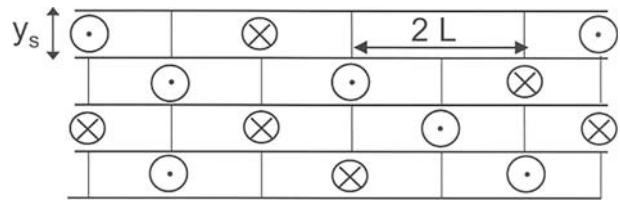


Fig. 10—Model of screw dislocations of both signs ($\odot$, $\otimes$), distributed in "boxes" of length $2L$ and height y_s , to simulate dislocation glide in PSB channels, as proposed in Ref. 56.

where ρ is the dislocation density, L the dislocation slip path, and b the modulus of the Burgers vector $\mathbf{b}$. It is then reasonable to assume that screw dislocations that glide to and fro by distances L will not be annihilated, unless they encounter a screw dislocation of opposite sign on a glide plane closer than y_s . This condition is fulfilled, when each screw dislocation is confined to a "box" of height y_s and length $2L$, as indicated in Figure 10, which shows screw dislocations of both signs (indicated by circles with a cross or a dot in the center) distributed in this manner. The criterion for annihilation can hence be formulated as

$$\rho \geq \frac{1}{2 \cdot L \cdot y_s} \quad [6]$$

or, after substitution of ρ from Eq. [5],

$$\beta = \frac{2\gamma_{pl}y_s}{b} \geq 1 \quad [7]$$

Similarly, annihilation should not occur for

$$\beta = \frac{2\gamma_{pl}y_s}{b} \leq 1 \quad [8]$$

Equations [6] and [7] thus are measures for the irreversibility of dislocation glide, whereas Eq. [8] applies to reversible dislocation glide. If now the known values for the local shear strain amplitude in PSBs, namely, $\gamma_{pl,PSB} \approx 10^{-2}$ ^[53] and $y_s \approx 50$ nm, are inserted, one finds that $\beta \approx 3$,^[56] indicating that cyclic slip in the PSBs is indeed highly irreversible. We note in passing that, had one assumed that dislocations glide in groups of n dislocations, then ρ in Eq. [6] would be larger by a factor n , whereas β in Eqs. [7] and [8] would be smaller by a factor n . Applying the same procedure to the matrix with the much lower local shear strain amplitude $\gamma_{pl,M} \approx 10^{-4}$, one obtains $\beta \approx 0.024$, implying that slip in the matrix is almost completely reversible. In view of the much more extensive dislocation glide in the PSBs, as compared to the matrix, this result is as expected.^[56]

Essmann has developed a detailed statistical model of glide of screw dislocations in PSBs.^[25] In this model, Essmann finds that the cyclic slip irreversibility in PSBs is $p_{PSB} \approx 0.4$. When also the glide of edge dislocations is taken into account, the cyclic slip irreversibility is lowered by about a factor of 2. Thus, $p_{PSB} \approx 0.2$ to 0.4. It is interesting to note that Woods,^[51] in a study of fatigued Cu-5 at. pct Al single crystals, estimated the slip irreversibility as $p \approx 0.3$ by assuming that the

softening observed in the approach to cyclic saturation was caused by the destruction of short-range order by random slip.

VI. EXTRUSIONS (AND INTRUSIONS) IN FATIGUED FCC METALS

A. EGM Model^[21,26,58,88]

Based on the detailed knowledge of the dislocation glide mechanisms in PSBs in cyclic saturation of fcc single crystals, as summarized previously, Essmann *et al.*^[88] developed a semiquantitative model (called EGM model subsequently) of the evolution of the surface relief of PSBs. Polák and co-workers have developed a related model, which is based on similar considerations as the EGM model but differs in the way in which diffusional effects occurring between PSB and matrix are taken into account.^[23,91] Both models are essentially based on the knowledge of the dislocation processes in fatigued copper single crystals. It is believed that the underlying assumptions are in certain ways so general that the predictions, perhaps with slight modifications, will also apply to other materials in which PSBs do not necessarily have the ladder structure.

Basic ingredients of the EGM model are that, in the PSBs, conditions of steady-state cyclic deformation prevail and that there is hence a balance between dislocation multiplication and annihilation for both screw and edge dislocations. As will be shown, it is the annihilation of the edge dislocations that will occur preferentially in the regions of high edge dislocation density, *i.e.*, in the walls, that is particularly important with regard to the evolution of the extrusions in the surface relief. When two edge dislocations on separate glide planes annihilate mutually, either vacancies or interstitials must be left over. In the case of the PSBs, there are, in addition to the observation that there is an excess of vacancy-type dipoles,^[92] a number of reasons why the net result of edge dipole formation and annihilation should lead to a *net excess* of some form of vacancies in cyclic saturation.^[88] When a PSB lamella is deformed homogeneously throughout its volume, then this is envisaged to occur through “chainlike” sequences of glide processes, terminated by the annihilation of the leading edge dislocations of two neighboring glide zones. This is illustrated for a single crystal in Figure 11 for a sequence of glide processes, which are inclined slightly to the glide plane, with annihilation of either vacancy or interstitial edge dipoles. In the former case, the slip steps at the surface lead to the formation of extrusions on both sides, whereas in the latter case, intrusions would be formed. In view of the predominance of vacancy-type edge dipoles, there should be a net extrusion effect. When one considers that the PSB lamellae must be filled homogeneously by the dislocation flux of stacks of neighboring “chains” of glide processes, which are inclined under an angle to the glide plane, then edge dislocations will be deposited at the ends of these chains at the PSB-matrix interfaces. Because of the dominance of vacancy-type edge dipoles,

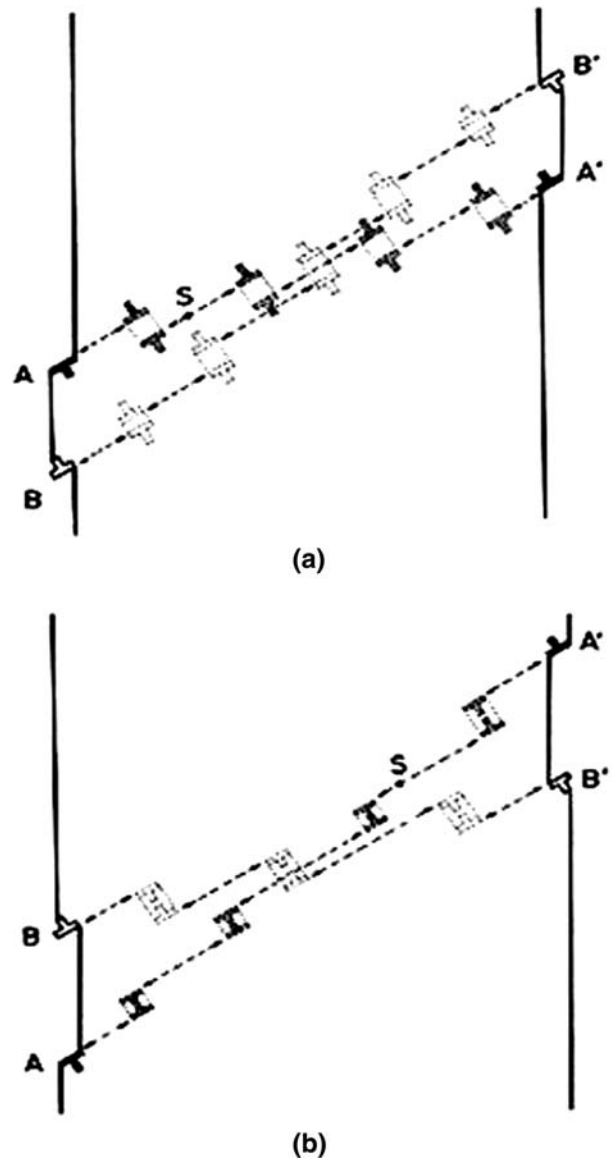


Fig. 11—Basic mechanisms of (a) extrusion formation, (b) intrusion formation, by chains of glide zones terminated by annihilation of dislocations, according to EGM model^[88]. For the sake of clarity, all the other dislocations (in the walls and channels) have been omitted.

these interface dislocations will have their extra half-planes inside the PSB, as shown in Figure 12. It is then easily seen that, in the tension half-cycle, the interface dislocations can and will glide out at *A* and *A'*, whereas in the compression half-cycle, the remaining interface dislocations can and will glide out at *B* and *B'*, leaving behind so-called “static” extrusions. In the EGM model, the height *e* of the static extrusions follows from the fact that, because of mass conservation, there will be a volume expansion of the PSB lamella, originating from the *local* vacancy concentration C_v^{sat} attained in cyclic saturation in the dislocation walls where the annihilations occur (volume fraction of walls $f_w \approx 0.1$). The local vacancy concentration C_v^{sat} could be estimated as 6×10^{-3} .^[21,58,87] Then, averaged over the entire PSB lamella, the mean vacancy concentration would be $f_w \cdot C_v^{\text{sat}} \approx 6 \times 10^{-4}$. The corresponding volume change

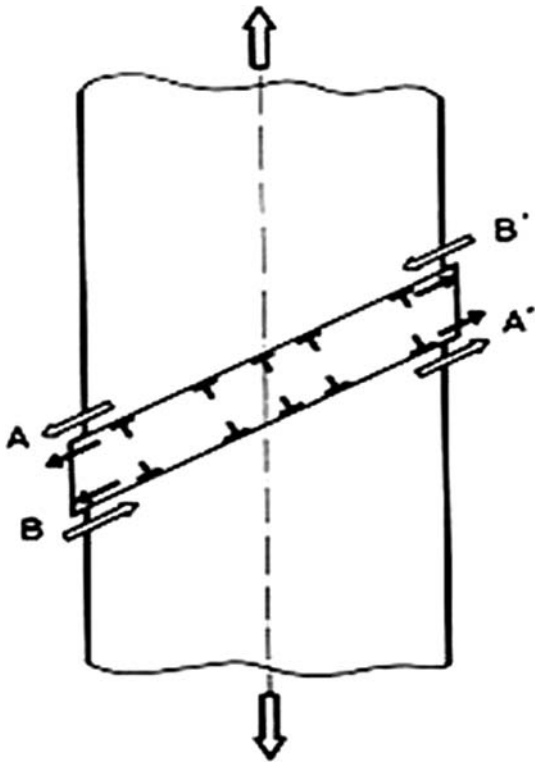


Fig. 12—Action of external and internal stresses on interface dislocations in PSB. Full arrows denote repulsive forces between dislocations, and open arrows denote forces caused by external stress.^[88]

of the PSB lamella comes about by a one-dimensional elongation of the PSB lamella in the direction of the Burgers vector $\mathbf{b}$ and leads to one static extrusion of height e on each side of the crystal:

$$e \approx 0.5 C_v^{\text{sat}} \cdot f_w \approx 3 \times 10^{-4} D \quad [9]$$

where D is the specimen diameter measured in the direction of the Burgers vector. In other words, it is predicted that the extrusion heights will be the larger, the thicker the fatigued single crystal is. In the case of a polycrystal, D would represent the grain size, and the predicted extrusion height would be twice as large as according to Eq. [9], because only one extrusion would form. Nonetheless, extrusions in polycrystals should be much smaller than in single crystals, because the grain size is always much smaller than the specimen diameter.

The described process of material extrusion is purely mechanical and can also operate at low temperatures, as observed.^[22,60,76,93,94] It is envisaged that, after the static extrusions have been formed rather rapidly and provided that diffusion can occur, there will be a slower continuous extrusion growth, since vacancies lost to the PSB by diffusion to the matrix are replaced by the glide and annihilation processes described, whereby additional PSB-matrix interfacial dislocations form. Furthermore, the continued cyclic deformation in the PSB will lead to a gradual roughening of the surface of the extruded material. According to Reference 26, the evolution of surface roughness in the PSB can be described by the mean width $\bar{w}$, defined as the largest

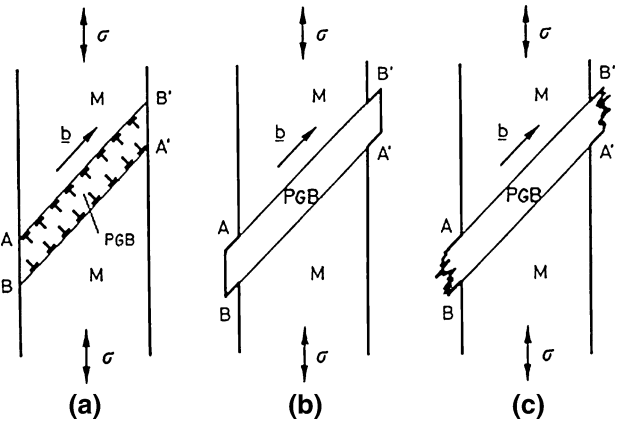


Fig. 13—Schematic illustration of evolution of surface relief at emerging PSB according to the EGM model: (a) interfacial dislocations accumulate at PSB-matrix interfaces, (b) rapid formation of static extrusions by emergence of interfacial dislocations, and (c) gradual roughening of surface of extrusions by random slip in PSB.^[21]

peak-to-valley spacing, measured parallel to the Burgers vector, as

$$\bar{w} \approx \sqrt{6bhpN\gamma_{pl,PSB}} \quad [10]$$

where h denotes the thickness of the glide region studied. With appropriate values for a PSB of thickness $1 \mu\text{m}$ in fatigued copper,^[26] this leads to

$$\bar{w} \approx 1.5 \times 10^{-3} \sqrt{N} (\mu\text{m}) \quad [11]$$

This surface roughening effect should be similar in polycrystals and in single crystals, whereas the heights of extrusions should be much smaller in polycrystals than in single crystals. In summary, the EGM model predicts the following sequence of events for fatigued mono- and polycrystals:

- (1) rapid formation of *static extrusions* with heights proportional to D , as defined previously;
- (2) gradual *surface roughening* of the extruded material; and
- (3) *continuously growing extrusions* at temperatures at which diffusional processes occur.

The surface profile predicted by the EGM model at lower temperatures at which continuous extrusion growth does not occur is illustrated schematically in Figure 13. It should be noted that, aside from the valleys in the surface roughness of older PSBs, the EGM model in its original form describes only the development of extrusions but not of intrusions. Polák^[23,91] modified the EGM model in some respects, in particular as follows. (1) Because of the diffusion of vacancies from the PSB into the adjacent matrix, intrusions should form in the matrix adjacent to the PSB-matrix interfaces. (2) Furthermore, continuous extrusion growth was postulated to occur over the entire width of the PSB, whereas in the EGM model, it was assumed that the diffusional processes would be confined to narrow regions near the PSB-matrix interface, giving rise to ribbonlike extrusion growth at the interfaces.^[88] The model of Brown^[19,84,95] is similar in several respects to the EGM

model. One important difference is that the interface dislocations (*cf.* Figure 12) have their extra half-plane not inside but outside the PSB. Brown does not consider the emergence of these interface dislocations (which would produce only intrusions) but instead analyzes the stress singularities at the PSB-matrix interface at the surface and the internal stresses induced by these dislocations and thus derives a crack initiation criterion.

In the EGM model, stage I cracks are expected to initiate transgranularly at the PSB-matrix interfaces. It is considered probable that these microcracks have, in the past, frequently been considered as “intrusions.” In addition, the possibility of crack initiation in the sharp valleys of the roughness profile cannot be excluded. In this context, mention should be made of the recent work of Risbet and Feaugas,^[41] who defined fatigue crack initiation in terms of a critical plastic strain irreversibility. Finally, there is the possibility of intergranular crack initiation by PSBs, which impinge against grain boundaries (GBs), with their Burgers vector directed at the GBs. The EGM model can explain the occurrence of such PSB-GB cracks. The latter have been observed frequently^[26,96,97] and are dominant at low and intermediate amplitudes.^[98]

B. More Recent Experimental Observations of the Surface Relief of Fatigued Fcc Metals

In the last 25 years, several groups have studied the surface reliefs of fatigued metals in more detail, making

use of novel techniques that had not been available in most of the very early investigations, such as SEM, contamination line profile analysis by SEM,^[26,58,98] AFM,^[40,41,69,70,99–102] and SEM applied to specimens sectioned with the so-called sharp corner technique developed by the Basinskis^[22,60,67,103] and subsequently applied also by Laird and co-workers^[59,104] or by the similar micromilling technique of Neumann's group.^[18,105]

In general, the surface profiles of PSBs observed by different authors have been found to be quite complex and to differ in details that depend on the type of fatigue testing and on the observational techniques employed. “Young” PSBs generally exhibit less complex surface reliefs. As an example, Figure 14 shows two views of the surface reliefs of a rather young PSB that had formed after about 500 cycles at $\gamma_{pl} = 5 \times 10^{-4}$.^[26] In this case, the so-called contamination line technique was used to delineate the elevations and depressions of the surface relief. The distortions of the contamination line indicate extrusions of “triangular” shape of a height of a little more than a micron in both Figures 14(a) and (b). Such triangular extrusions are predicted by the EGM model in the case of PSBs that do not extend deep into the bulk but “peter out.” In addition to the extrusion, a much smaller “intrusion” is apparent at the side of the extrusion shown in Figure 14(b), whereas the extrusion shown in Figure 14(a) does not exhibit such an intrusion. In several cases, one and the same PSB could be followed all around a fatigued single crystal, and it

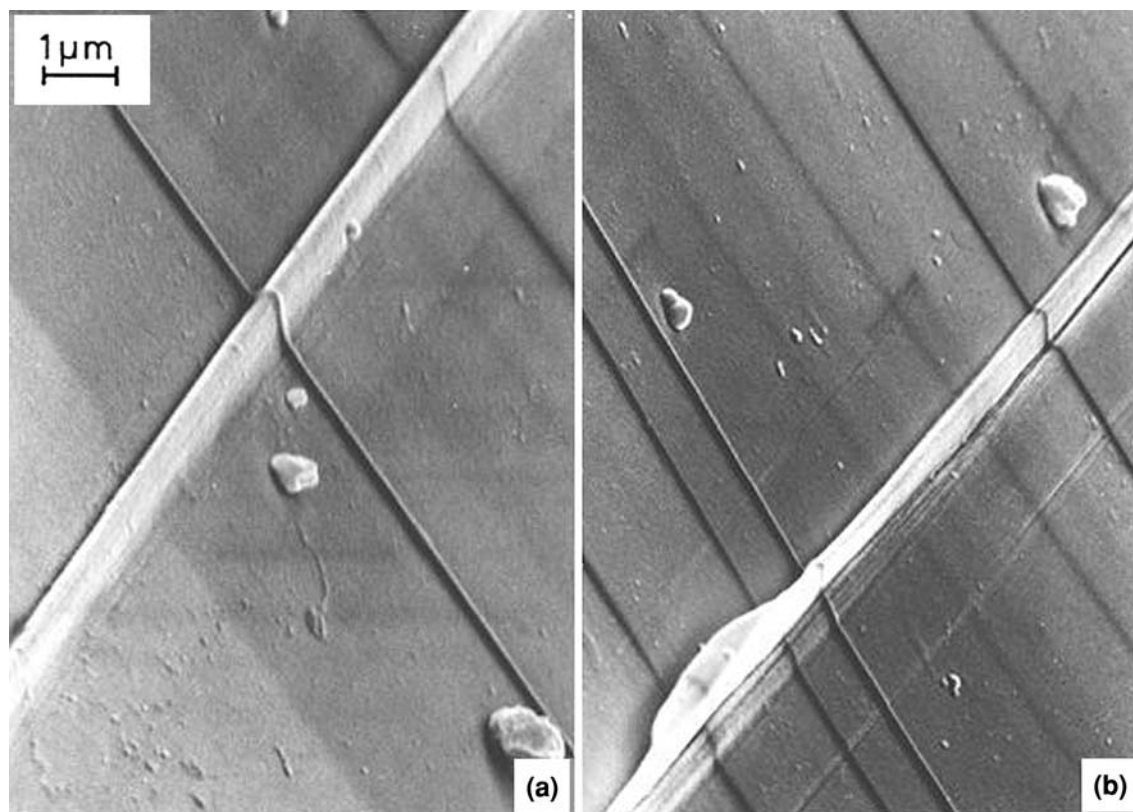


Fig. 14—Young PSBs in fatigued copper, age approximately 1000 cycles, imaged in SEM by the contamination line technique: (a) rounded triangular shaped extrusion; and (b) other part of same PSB as in (a), with ribbonlike extrusion growth at bottom right and development of small intrusion.^[26]

could be confirmed that there were extrusions of similar height on both sides of the crystal, proving that the PSB slab had experienced an overall volume increase. Thus, extrusions seem dominant, whereas the intrusion effect is not only less pronounced, but also less frequent and sometimes completely absent. The SEM micrograph in Figure 15 shows the surface profiles of older PSBs in a view on the top face of a copper single crystal that had been fatigued to failure ($N_f = 1.34 \times 10^5$ cycles) at $\gamma_{pl} = 2 \times 10^{-3}$. In addition to the complex pronounced extrusions, stage I microcracks are revealed at the PSB-matrix interfaces at sites corresponding to A and A' in Figure 12. Comparison of older PSBs formed at different temperatures reveals some characteristic differences,

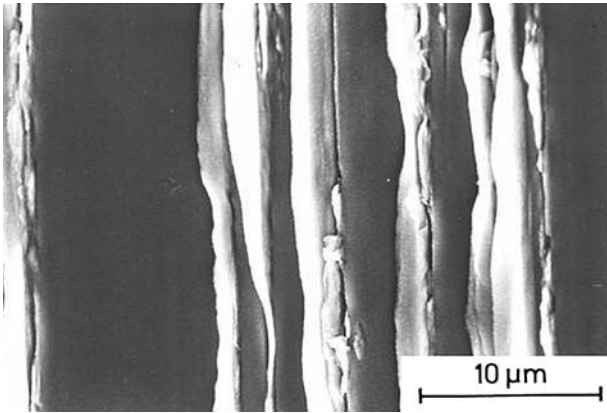


Fig. 15—SEM view on top face of fatigued copper single crystal after fatigue to failure ($\gamma_{pl} = 2 \times 10^{-3}$, $N_f = 1.34 \times 10^5$ cycles). Old extrusions with stage I shear cracks at PSB-matrix interfaces.^[26]

as shown in Figure 16. After fatigue at lower temperatures (Figure 16(a)), the extrusions are bulgy and have an extrusion height of approximately $2 \mu\text{m}$, which compares favorably with the predicted static extrusion height (for $D \approx 6 \text{ mm}$). In addition, the surface of the extrusion exhibits the roughening effect proposed. After fatigue at a higher temperature (Figure 16(b)), the extrusions are narrower, have a more ribbonlike shape, and have a somewhat larger height. The latter features indicate that some continuous extrusion growth has occurred.

Related studies on fatigued copper polycrystals^[26,58,106] fatigued at room temperature confirmed that the extrusion heights were much smaller than in single crystals and confirmed the occurrence of surface roughening of the extrusions. Thus, for example, for room-temperature fatigue at $\Delta\epsilon_{pl} = 2 \times 10^{-3}$, $N = 3 \times 10^4$, a mean profile width $\bar{w} \approx 0.4 \mu\text{m}$ was obtained^[26] in good semiquantitative agreement with the predictions according to Eqs. [9] and [11]. It was noted that the appearance of the extrusion/intrusion profiles varied with the orientation of the grains.^[106] Measurements of the evolution of extrusions as a function of numbers of cycles at different temperatures yielded the interesting result that, at 77 K, extrusion growth ceased completely, whereas continuous extrusion growth was observed after fatigue at higher temperatures,^[58] as shown in Figure 17. This confirms the expectation that, in the absence of diffusion, only static extrusions develop and that their growth ceases, when all interface dislocations have emerged, as discussed previously. In recent years, several groups have performed detailed studies of the surface relief of emerging slip bands by AFM. In these studies, the surface reliefs of fatigued

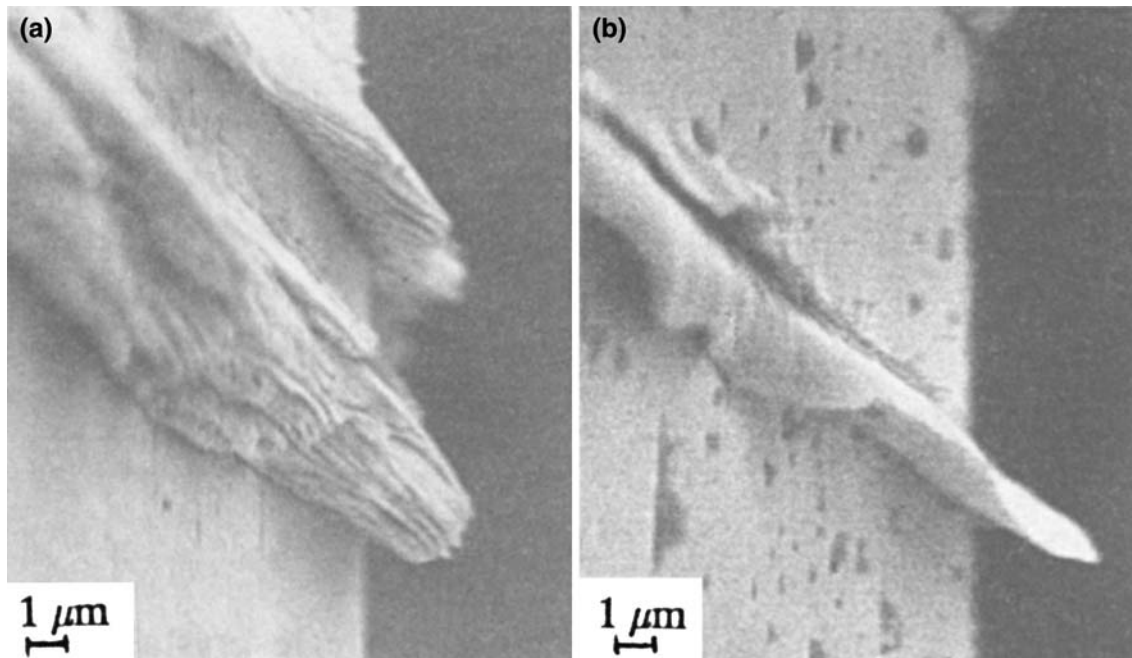


Fig. 16—Older extrusions (age about 40,000 cycles) in copper single crystals fatigued at $\gamma_{pl} = 2 \times 10^{-3}$: (a) after fatigue at 77 K and (b) after fatigue at 403 K. Stress axis is vertical.^[58]

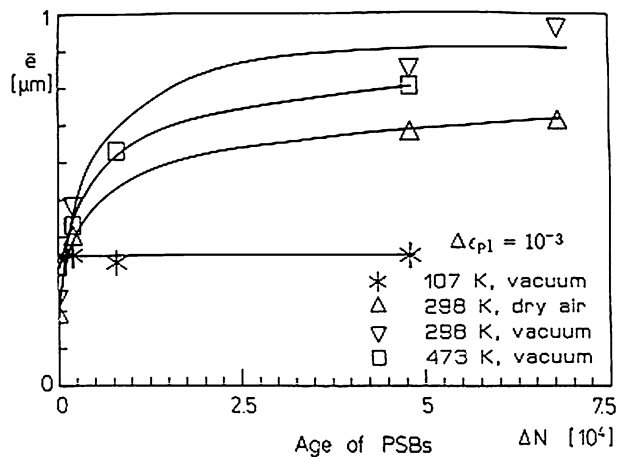


Fig. 17—Average extrusion heights $\bar{e}$ in copper polycrystals fatigued at different temperatures, plotted as a function of numbers of cycles N .^[58]

polycrystalline copper,^[39] austenitic stainless steel,^[99–102] and ferritic stainless steels^[61] were investigated. In all cases, impressive evidence of slip band extrusions was obtained. As expected for polycrystals, the extrusion heights were rather small, typically well below 1 μm , but could nonetheless be resolved reliably by AFM. The limitations of AFM lie not in inadequate resolution but in the fact that only parts of the surface relief can be traced by the AFM tip.^[61,99–101] Because the extrusions emerge under an angle of approximately 45 deg to the surface, only the contours of the extrusions on the sides that make an obtuse angle with the surface are accessible to AFM. In addition, narrow intrusions/cracks cannot be studied, because the AFM tip cannot enter into narrow depressions. This latter problem can be overcome by studying replicas of the surface by AFM, as illustrated in Figure 18 from the work of Polák *et al.*^[61] Figures 18(a) and (b) illustrate schematically for an extrusion with adjacent intrusions how the AFM tip can follow the reliefs of the original surface and of the replica, respectively. Thus, by combining AFM and the other techniques, a rather reliable picture of the surface relief contours has been obtained in several cases. As an example, Figure 19 gives a good impression of the surface relief in grains of fatigued stainless steel 316. The dependence of the surface relief on temperature was studied comprehensively by Basinski and Basinski^[22,60] on copper single crystals fatigued at temperatures between 4.2 and 350 K. Broadly speaking, the most important finding was perhaps that, after fatigue at temperatures below room temperature, mainly extrusions were observed, which were always bulgy, whereas after fatigue at higher temperatures, “nonbulged” extrusions and intrusions were also found. Figure 20 shows an example of a bulgy extrusion with a stage I crack at the PSB-matrix interface, formed during fatigue at 115 K. Basinski and Basinski^[22] have given a very detailed and balanced description of the observations of the surface reliefs of different fatigued materials, as studied by different authors who used a variety of imaging techniques. In spite of many similar results,

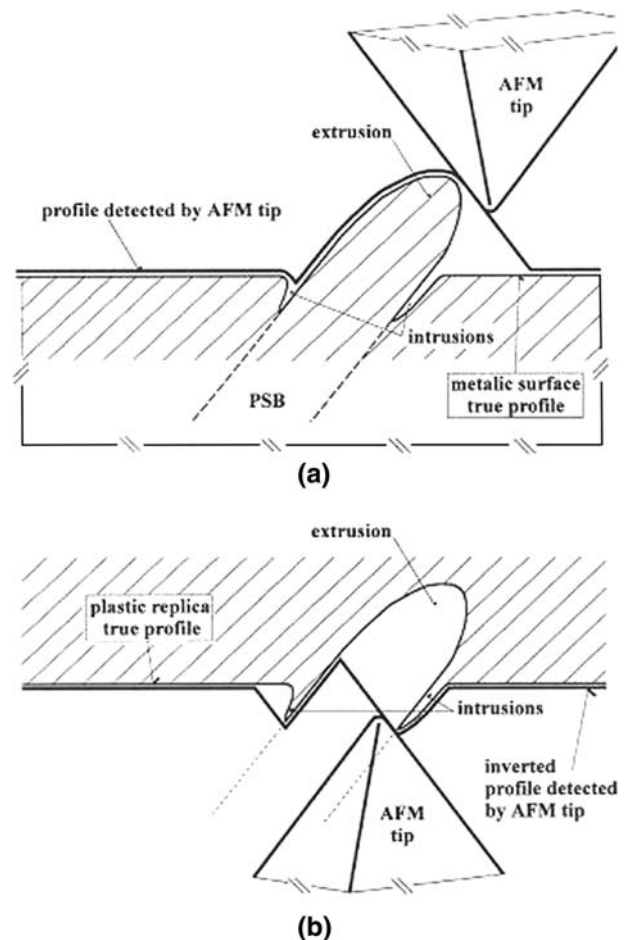


Fig. 18—Assessment of surface relief of fatigued metals by AFM. (a) AFM of surface of fatigued specimen, detection of extrusions. (b) AFM of surface replica of fatigued specimen, detection of intrusions.^[61] Courtesy of the authors.

some discrepancies have also been reported. While some of the discrepancies may be due to semantic problems, others seem real and have not been fully resolved. In spite of differences in the details of observation and interpretation, it is fair to conclude that the studies show agreement in some main features, which can be summarized as follows. In general, in particular at lower temperatures, extrusions are frequently bulgy and dominant, whereas intrusions are not always observed. Even when intrusions appear, they do not necessarily appear paired with extrusions. This confirms earlier doubts about extrusion-intrusion pairs being representative.^[15,21,24,26,88,107] In many studies at room temperature, it was clearly observed that, at first, the surface relief consisted only of extrusions. Very narrow intrusion-like deepenings, preferentially at the PSB interfaces, appeared only at a later stage. In several instances, the morphology of these intrusions was quite different from that of the extrusions, resembling actually narrow microcracks more than “valleys” of similar width as the extrusion “hillocks.” Hence, intrusions, when found, can in fact frequently be considered as embryonic stage I shear cracks at the PSB-matrix interface rather than as “inverse” extrusions. Thus, regarding the main features,

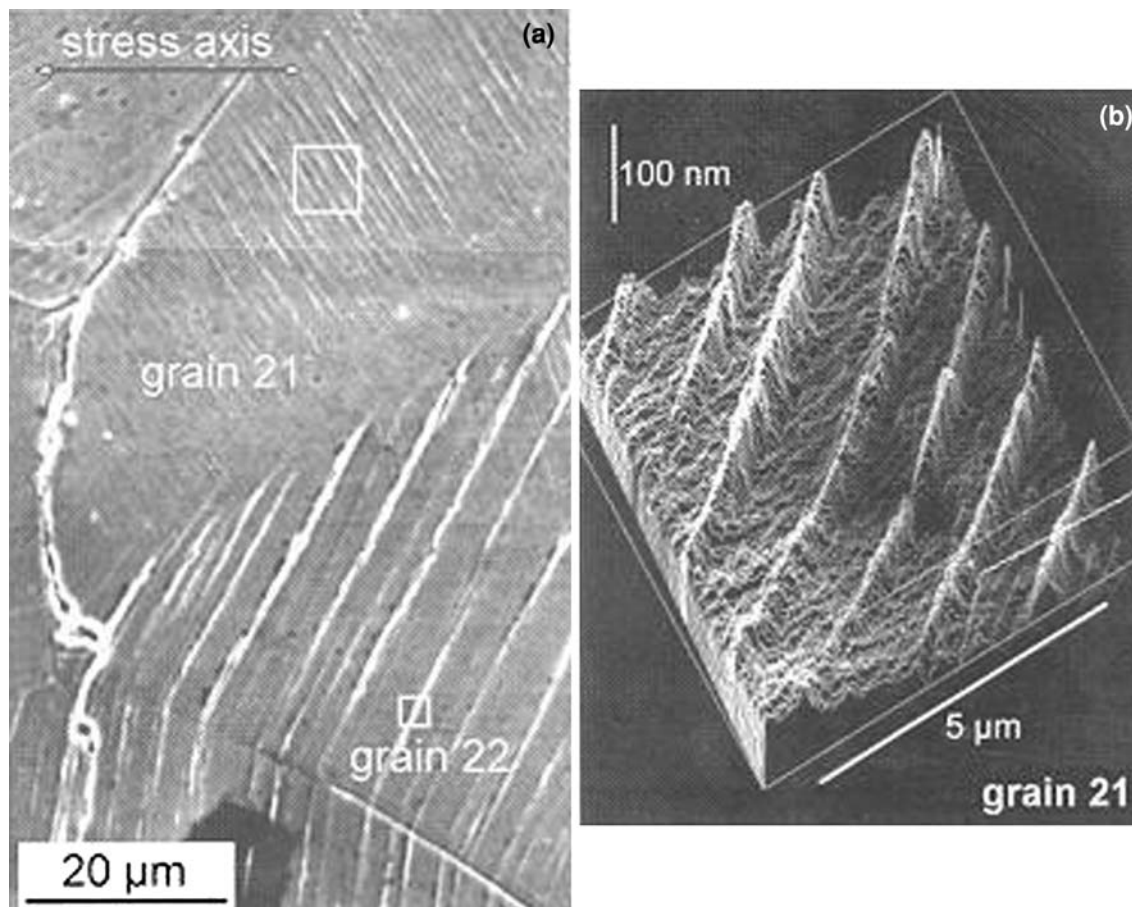


Fig. 19—AFM images of surface relief of fatigued austenitic stainless steel 316 L. (a) Survey of different grains. (b) Enlarged view of area in grain 21.^[61] Courtesy of J. Polák.

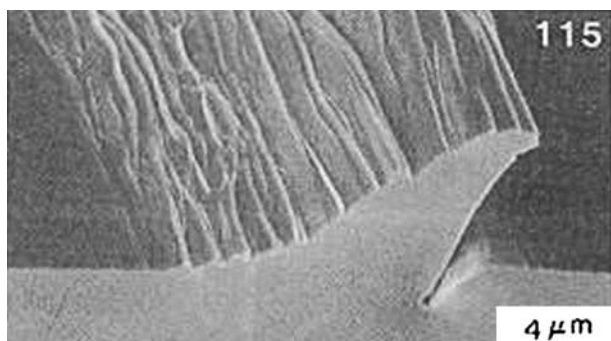


Fig. 20—Sharp-corner SEM image of bulky extrusion with adjacent stage I shear crack in copper single crystals fatigued at 115 K.^[60] Courtesy of S. Basinski.

the agreement with the EGM model^[26,58,88] and Polák's model^[23,61,101] is fair.

VII. SURFACE RELIEF AND CYCLIC SLIP IRREVERSIBILITY IN THE ULTRA-HIGH-CYCLE FATIGUE REGIME

The preceding considerations concerned mainly the processes occurring in the high-cycle fatigue (HCF)

range for fatigue lives of typically 10^5 to 10^6 cycles to failure. In recent years, there has been a growing interest in the regime of much longer fatigue lives of 10^9 or more cycles to failure (compare the proceedings of the last conference on very-high-cycle fatigue (VHCF)).^[108] It has been found that, in the range of so-called VHCF or ultra-high-cycle fatigue (UHCF), fatigue failures frequently occurred even when the loading amplitude was believed to lie in the safe range below the fatigue limit. Here, an example of recent work on the UHCF of copper polycrystals is presented.^[34,35] The goal of the study was to see whether fatigue damage would still develop, when the cyclic loading was carried out at loading amplitudes below the traditional PSB threshold and extended into the UHCF range. Under normal circumstances, fatigue damage would not have been expected to occur under these circumstances. Nonetheless, in ultrasonic fatigue tests performed up to more than 10^{10} cycles well into the UHCF regime, it was found that cyclic strain localization occurred in lamellar bands, which were shown to be persistent. In addition, a marked surface roughness developed, and stage I shear cracks initiated at the sites of stress concentrations. This behavior shows that cyclic deformation below the PSB threshold still retains a small irreversible component, which, accumulated over a very large number of cycles,

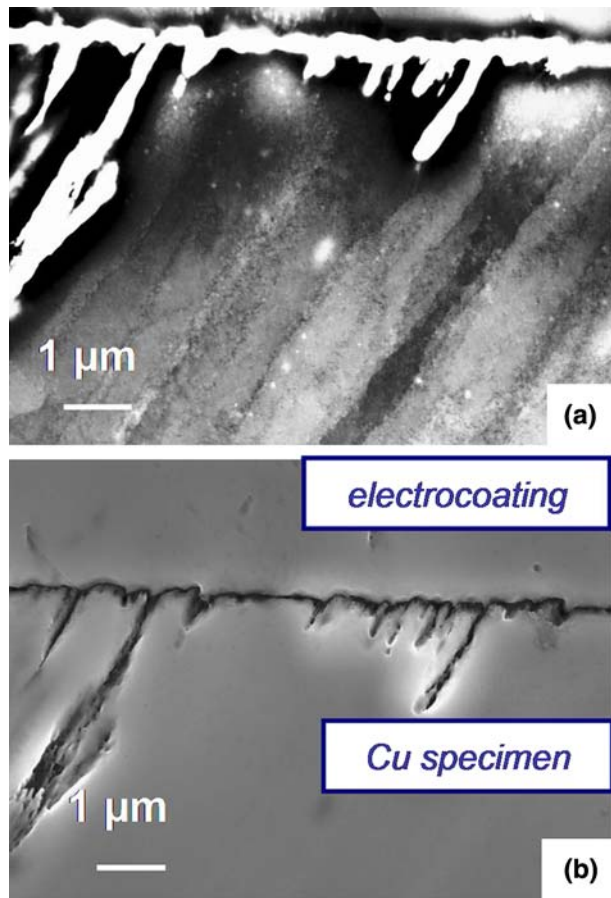


Fig. 21—SEM images of section parallel to axis of copper polycrystal after ultrasonic fatigue for 1.59×10^{10} cycles, 6 MPa below the traditional PSB threshold. (a) SEM image with BSE detector, showing lamellae of strain localization. (b) SEM image of same site with in-lens SE detector showing surface roughness more sharply and stage I shear cracks.^[34,35]

can still lead to a very large *cumulative* irreversible plastic strain^[30] (cf. Eq. [5]). The details are documented in Figures 21 and 22, which show sections parallel to the axis of a fatigued copper polycrystal that had been fatigued up to 1.59×10^{10} cycles. In order to protect the surface, the specimen had been electrocoated with copper before sectioning. In the SEM micrographs shown in Figure 21, the surface relief can be recognized as the interface between the fatigued specimen and the electrocoating. In Figure 21(a), taken with a backscattered electron (BSE) detector, the electron channeling contrast reveals lamellae of strain localization under an angle to the stress axis. The surface relief is not imaged very clearly. However, in the SEM image of the same area, taken with an in-lens secondary electron detector (Figure 21(b)), the surface relief and the stage I shear cracks are clearly recognizable. The surface roughness can be analyzed in more detail to yield a quantitative value for the cyclic slip irreversibility p as follows. Figure 22 shows an example at higher magnification of the surface roughness observed on a section with the free surface, which was generated by ion milling in a focused ion beam (FIB) instrument. The original surface had

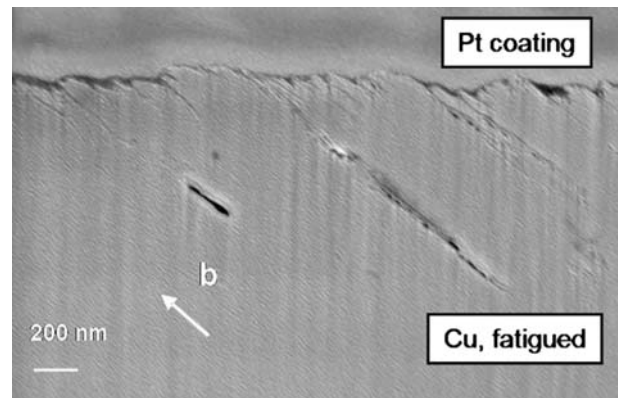


Fig. 22—Higher magnification SEM image of surface roughness of copper polycrystal after ultrasonic fatigue for 1.59×10^{10} cycles, 2 MPa below the traditional PSB threshold. The text provides details.^[34,35]

been coated in the FIB with a platinum layer of approximately 2-μm thickness. The roughness can be characterized by its mean width $\bar{w} \approx \sqrt{6bhpN\gamma_{pl}}$, as described earlier in Eq. [10]. Analysis of the mean width $\bar{w}$ over a distance $h = 2 \mu\text{m}$ yielded a value $\bar{w} \approx 150 \text{ nm}$. Inserting the value of 1.37×10^{-5} for the plastic shear strain amplitude γ_{pl} at the PSB threshold,^[109,110] all quantities in Eq. [10] are known except for the cyclic slip irreversibility p . Resolving the equation for p , the value $p = 0.000034$ is obtained.^[34,35] This result implies that p is indeed very small and in fact almost negligible. However, it must be realized that the very small irreversible shear strain per cycle, accumulated over a very large number of cycles, sums to a sizeable value of the cumulative irreversible shear strain:

$$\gamma_{pl,irr,cum} = 4N \times p \times \gamma_{pl} \approx 34 \quad [12]$$

When considering that the cumulative irreversible plastic shear strain is as large as 34 (i.e., 3400 pct), it is not surprising that this cumulative effect led to significant microstructural changes and also to fatigue damage at the surface, in spite of the fact that the loading amplitudes were significantly lower than the traditional HCF PSB thresholds.

VIII. CYCLIC STRAIN LOCALIZATION AND SLIP IRREVERSIBILITY IN SOME ALLOYS

A. Fatigued Single-Phase Alloys

Several groups have investigated fatigued mono- and polycrystals of some simple fcc solid-solution-hardened alloys. One aim of these studies was to vary the slip character by alloying in order to investigate characteristic features of and differences between wavy and planar slip materials. The work of Lukáš and Klesnil^[111,112] on fatigued Cu-30 at. pct Zn single crystals is an interesting example of a study on a fatigued planar-slip material, which deforms by glide of dislocation groups. By electrocoating and sectioning the

fatigued specimens, the authors could image the surface roughness, as shown in one case in Figure 23. As shown in Reference 26, similar roughness profiles could be obtained by computer simulations using appropriate parameters.

The result obtained in these simulations for the example shown in Figure 23(a) is presented in Figure 23(b). Then, with Eq. [10], and inserting the experimental test data, the cyclic slip irreversibility p could be obtained. The following results were obtained for two cases investigated: (1) $\Delta\sigma/2 = 82$ MPa, $N = 5 \times 10^5$ cycles: $p \approx 0.5$ pct, and (b) $\Delta\sigma/2 = 118$ MPa, $N = 2 \times 10^4$ cycles: $p \approx 3$ pct. Although there is some uncertainty about the accuracy of this method to obtain quantitative values of the cyclic slip irreversibility p , the finding that p decreases, *i.e.*, that slip becomes more reversible, as the loading amplitude is reduced (and the

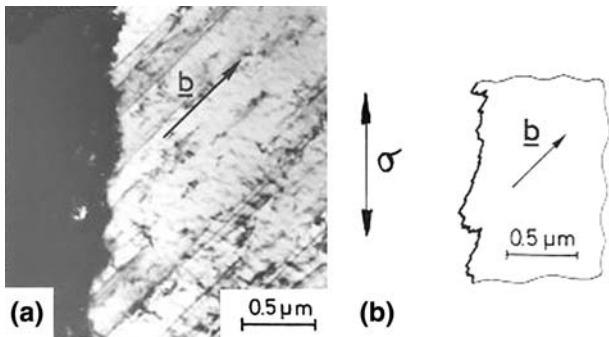


Fig. 23—Fatigue-induced surface roughening in α -brass single crystal. (a) TEM of electrocoated specimen sectioned parallel to stress axis. Burgers vector $\mathbf{b}$ lies in plane of section.^[111,112] Courtesy of the authors. (b) Computer simulation of fatigue-induced surface roughness.^[26]

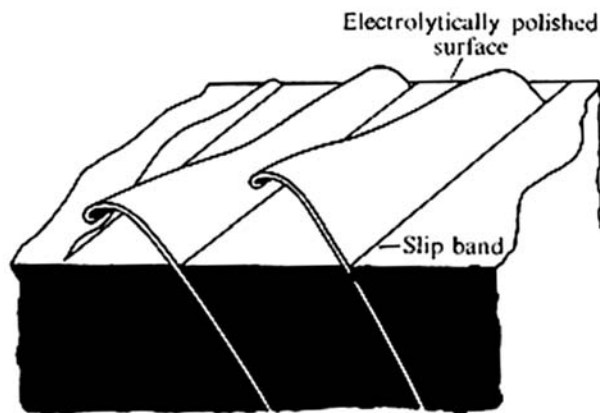
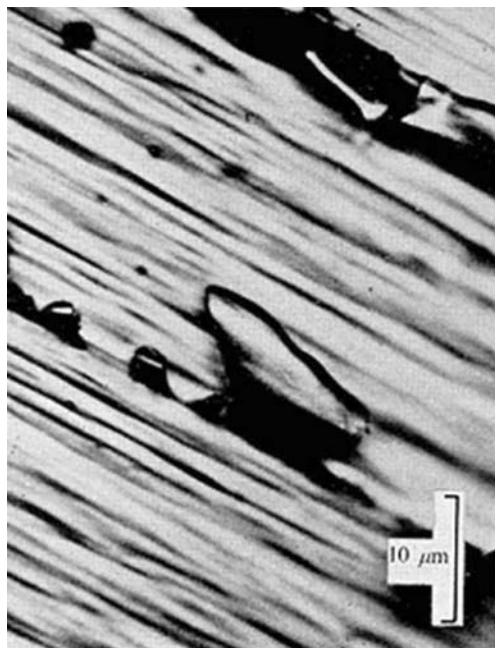


Fig. 24—Slip band extrusions in fatigued age-hardened Al-4 pct Cu alloy.^[73]

fatigue life increases) is according to expectation and seems very realistic.

B. Fatigued Precipitation-Hardened Alloys

The most severe cyclic strain localization is observed in precipitation-hardened alloys containing shearable precipitates. The PSBs are thinner ($0.1 \mu\text{m}$ or less) than in single-phase materials, and strain localization is more severe by at least one order of magnitude (*cf.*, References 17 and 113). As a consequence, the extrusion effect can be very large, as first discovered by Forsyth,^[73] considerably exceeding that in single-phase materials. The famous optical micrograph from the work of Forsyth and the schematic illustration of the effect are shown in Figure 24.

The appearance of PSBs in precipitation-hardened alloys containing shearable particles can be quite different, depending on the size of the particles and on the cyclic slip mode. This is illustrated schematically in Figure 25.^[21] In a simplified manner, one can distinguish between materials with wavy slip and rather small particles and materials with planar slip and rather large particles. The fatigue of underaged Cu-2 at. pct Co single crystals with spherical coherent pure cobalt particles, approximately 10 nm diameter, is a good example for the former case. As shown by Steiner *et al.*,^[114,115] PSBs develop in this alloy in thin lamellae in which the particles are cut repeatedly and chopped into smaller and smaller slices that are no longer recognizable in high-resolution TEM micrographs.^[27] This process is accompanied by severe cyclic softening, indicating that finally the Co atoms revert into solid solution. A simple criterion for cyclic softening can be formulated by assuming that, on average, the root-mean-square irreversible displacement $\sqrt{\langle x^2 \rangle}$ between

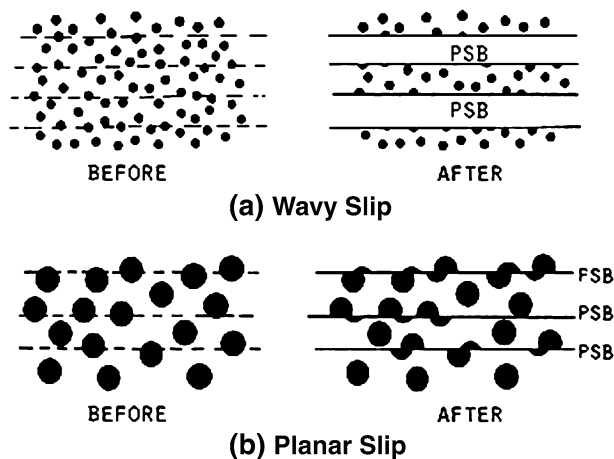


Fig. 25—Schematic illustration of types of PSBs in precipitation-hardened alloys containing shearable precipitates. (a) PSBs in wavy slip alloy with small coherent particles. PSB thickness is larger than particle size. (b) Very sharp PSBs in planar slip alloy with rather large particles. PSB thickness is much smaller than particle size.^[21]

neighboring atomic planes in the PSB must exceed the particle diameter d .^[27]

$$\sqrt{\langle x^2 \rangle} \approx \sqrt{4Nbh p \gamma_{pl,PSB}} > d \quad [13]$$

Comparing Eq. [13] with Eq. [10], it should be noted that the constant 6 under the square root has now been replaced by 4, and h now represents the atomic plane spacing.^[28,29] Then, from experiment, an average value for the *local* shear strain amplitude $\gamma_{pl,PSB}$ and the number of cycles N until softening is complete can be determined. Thus, the cyclic slip irreversibility p in the PSBs can be estimated. As discussed earlier, with an average value of 0.25 for the local shear strain amplitude and $N \approx 2000$ cycles, as deduced from the work of Steiner *et al.*,^[114,115] it is concluded that $p \approx 1$,^[27] i.e., slip in the PSBs is practically completely irreversible.

Nickel-base superalloys hardened by coherent shearable γ' particles are examples of planar-slip precipitation-hardened alloys with rather large particles (some $0.1 \mu\text{m}$ in size). Figure 26 shows, as an example from the work of Clavel and Pineau,^[116] a TEM micrograph of the alloy Waspaloy after fatigue to failure at 650°C . This alloy contained a bimodal γ' precipitate size distribution. In the figure, the Burgers vector lies almost exactly horizontally in the plane of the figure. Because the particles are easily resolved by TEM, the slip offsets stemming from the cutting of the particles during fatigue can be recognized clearly. Thus, it is possible to simply construct the sequence of all unreversed slip displacements, as indicated on the right side of the figure. Since the cumulative plastic shear strain of the alloy is known from experiment, the fraction of irreversible plastic strain, i.e., the cyclic slip irreversibility p , can be estimated and is found to be $p \approx 5$ pct.^[27] The two examples demonstrate again that microstructural fatigue damage is intimately related to the cyclic slip irreversibility.

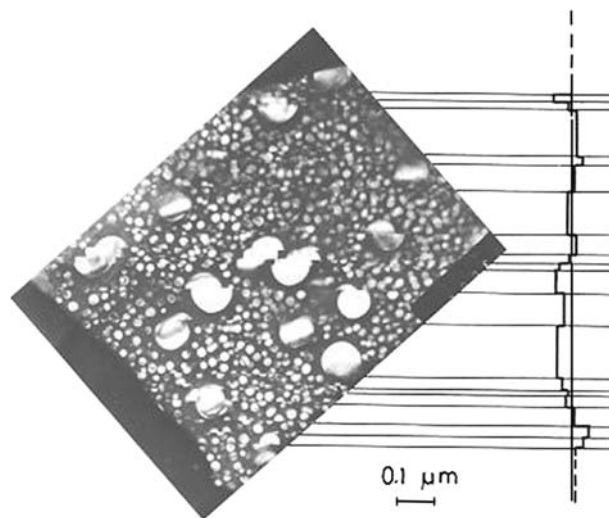


Fig. 26—Planar-slip PSBs in Waspaloy fatigued until failure at axial plastic strain amplitude $\Delta\epsilon_{pl}/2 = 2 \times 10^{-3}$, $N_f = 550$ cycles, and $T = 650^\circ\text{C}$. Taken from Clavel and Pineau,^[116] as presented in Ref. 27. Courtesy of the authors.

IX. FATIGUE OF BCC METALS

A. Summary of Specific Features of Dislocation Glide in Bcc Metals

Before presenting examples of fatigued bcc metals, it is expedient to draw attention to some features of the dislocation behavior of bcc metals, which differ from those of fcc metals as follows:

- (1) screw dislocations are immobile at low temperatures, giving rise to
- (2) high low-temperature yield stress, and to
- (3) “slip plane asymmetry,” meaning that glide of screw dislocations occur on different glide planes in forward and reverse directions at low temperatures.

These differences are related to the fact that screw dislocations in bcc metals have an “extended core” structure,^[117,118] making them sessile in their equilibrium configuration. Screw dislocations only become glissile under the action of an external stress and then glide by thermally activated spreading of double kinks. Thus, because rather high stresses are needed at low temperatures to make the rather immobile screw dislocations glide, the yield stress is high. At higher temperatures (typically above room temperature for metals such as α -iron and niobium), enough thermal activation is available, screw dislocations can glide easily, and the yield stress becomes much lower. This behaviour is illustrated in Figure 27. As indicated in Figure 27(a), the yield stress $\sigma(T, \dot{\epsilon})$, which depends on temperature T and strain rate $\dot{\epsilon}$, can be written as the sum of an athermal stress σ_G and a thermal stress $\sigma^*(T, \dot{\epsilon})$. The latter vanishes above the temperature $T \approx T_0$, which is shifted to higher temperatures at higher strain rates. Accordingly, the dislocation glide mode differs below and above $T = T_0$. Below T_0 , as shown schematically in Figure 27(b), the velocity $v_\odot$ of screw dislocations is much smaller than the velocity $v_\perp$ of edge dislocations.

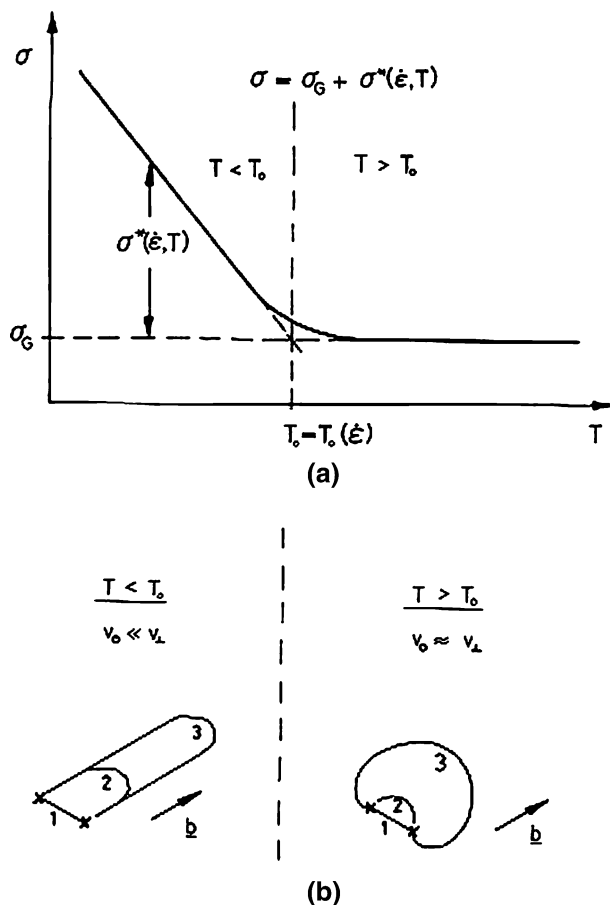


Fig. 27—Temperature dependence of mechanical behavior of bcc metals: (a) flow stress and (b) dislocation glide.^[21,119]

For this reason, dislocation glide occurs mainly by glide of edge dislocations, which draw out long straight (immobile) screw dislocation segments. Above T_0 , the mobilities of edge and screw dislocations become comparable, and dislocation glide becomes similar to that in fcc metals. The “slip plane asymmetry,” being also related to extended screw dislocation core structure, is also typical of the “low-temperature” behavior.

B. Dislocation Arrangements and PSBs in Fatigued Bcc Metals

All the named features make the cyclic deformation and fatigue behavior of bcc metals differ from that of fcc metals in the “low-temperature” regime.^[21,55,56,119–121] Because the temperature T_0 lies around room temperature for bcc metals such as α -iron, tantalum, and niobium and because the strain rates in fatigue are usually rather high (10^{-3} to 10^{-2} s $^{-1}$), these metals show typical features of low-temperature behavior at room temperature. Important consequences, as demonstrated in particular on fatigued α -iron single crystals,^[119–121] are as follows.

In the *cyclic microstrain* regime, at low plastic strain amplitudes, below about 5×10^{-4} , there is almost no cyclic hardening,^[21,119–121] because cyclic slip *in the bulk* can occur almost completely by the quasi-reversible to-and-fro glide of the mobile edge segments. Multiplication of dislocations is largely suppressed by the immobility of the drawn out screw segments; hence, only few dislocations are found. This situation is illustrated in Figure 28(a). However, *at the free surface*, emerging edge segments are lost. Cyclic plasticity is maintained by the local onset of multiple slip, which

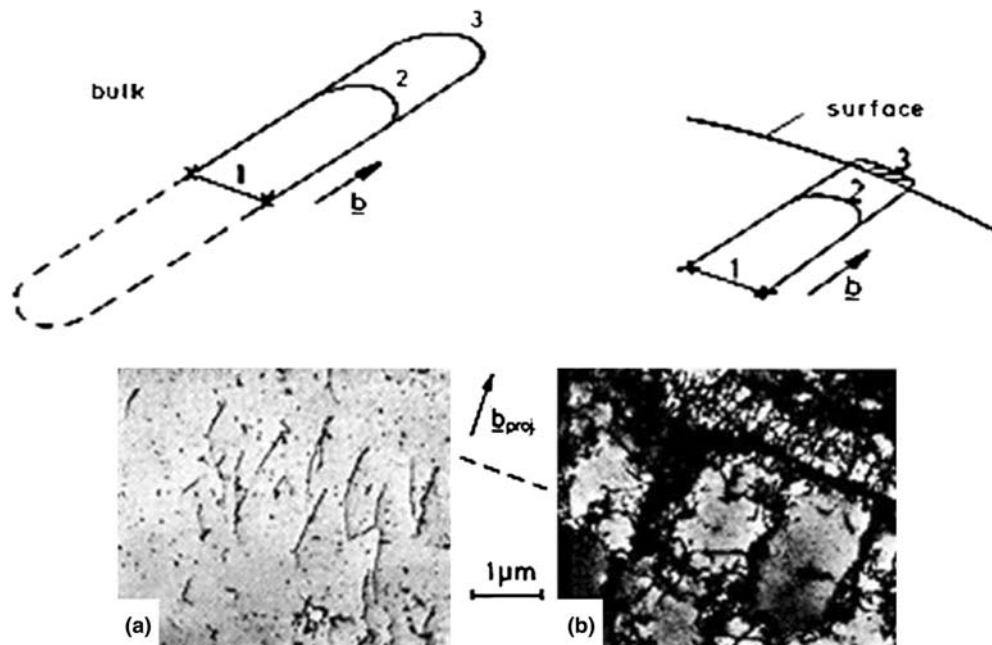


Fig. 28—Cyclic microyielding of bcc metals at low amplitudes: Dislocation glide modes, dislocation distributions in niobium single crystal after low-amplitude fatigue: (a) in the bulk and (b) near the free surface.^[21,119,122]

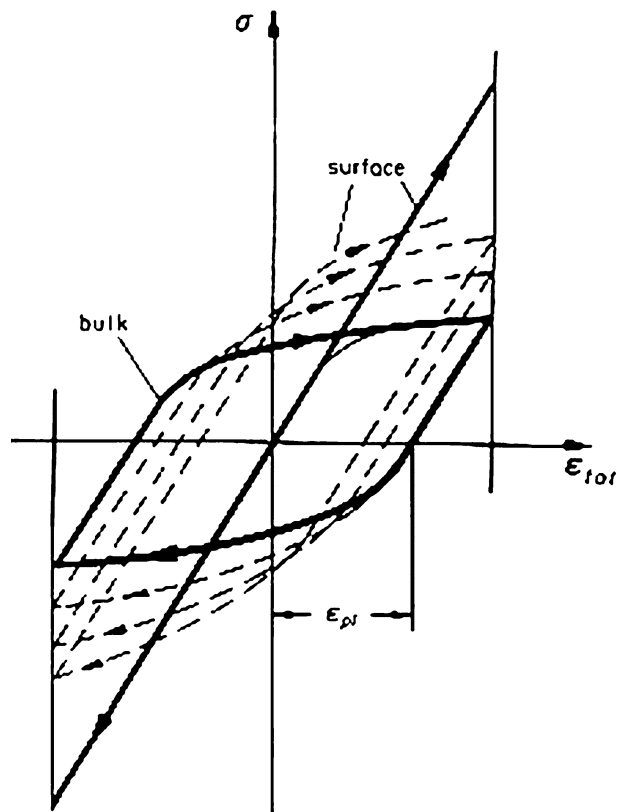


Fig. 29—Cyclic microyielding of bcc metals at low amplitudes. Schematic illustration of hysteresis loops for bulk and interior regions, showing preferential surface hardening.^[21]

leads to the formation of a cell structure,^[21,122] as shown in Figure 28(b). Similar observations have been made by Russian scientists on α -iron^[123] and molybdenum.^[124] Thus, cyclic hardening does occur, but is limited to a thin surface layer. These different cyclic hardening behaviors of bulk and near-surface regions are illustrated in plots of the hysteresis loops in Figure 29. In a strain-controlled test, cyclic surface hardening could at best continue until the local hysteresis loop of the near-surface region degenerates into an elastic line. Then, unless fatal cracks have already developed at that stage, no further fatigue damage should be able to develop. In the *cyclic macroyielding* regime at higher plastic strain amplitudes, bulk cyclic hardening will occur, with increasing participation of the screw dislocations, dislocation multiplication, and dislocation cell formation.^[21,56,119,120,122]

The PSBs similar to those in fatigued fcc metals, which exhibit distinct dislocation arrangements in the bulk, have not been observed in fatigued pure bcc metals. In view of what was said previously, this is not surprising, since dislocation glide occurs either quasi-reversibly at low amplitudes or by multiple slip with dislocation cell formation at higher amplitudes.^[21,56,119,120] Therefore, a matrix structure of veins of dipolar edge dislocation loops from which PSBs would originate cannot develop. Nonetheless, slip bands form at the surface with some roughness in which cracks

can initiate. In less pure bcc metals, the situation is different. In particular, the presence of interstitial carbon atoms modifies the behavior strongly due to dynamic strain aging. As shown by Wilson and co-workers^[126] (also Reference 127), real PSBs, although not with the ladder structure but in the form of some kind of “open” cell structure in dislocation-poor channels, form in low carbon steels, in particular at slightly elevated temperature at which the dislocation glide mode becomes rather similar to that in fcc metals. Similar observations have been made in α -iron single crystals containing small amounts of carbon, when deformed cyclically at low strain rates at room temperature.^[21,56,119] The dislocation distribution in the PSBs differs from that found in fcc metals, consisting essentially of dislocation-poor channels, which traverse a matrix of veinlike dislocation clusters. One reason for these observations is that the test conditions favor a behavior as expected for $T > T_0$. However, in addition, it is considered important that, in the presence of dynamic strain aging, the motion of the edge dislocations is retarded preferentially by the carbon atoms so that the dislocation velocities $v_{\parallel}$ and $v_{\perp}$ become more comparable, as in fcc metals.^[21,55,119] In only one study, PSBs with the ladder structure have been found in a fatigued low-carbon steel.^[128] It has remained unclear which circumstances were responsible for this exceptional behavior.

C. Cyclic Slip Irreversibilities and Fatigue Damage Related to the “Slip Plane Asymmetry”

As the contribution of the screw dislocations to cyclic deformation increases in the transition from cyclic micro- to macroyielding, the effect of slip plane asymmetry becomes more and more important, with dramatic consequences. In the case of single crystals of, e.g., α -iron^[21,121] or niobium,^[129,130] the slip plane asymmetry manifests itself in marked shape changes. Thus, the cross section of initially round cylindrical specimens becomes increasingly elliptical during push-pull cyclic deformation at room temperature. Since these shape changes result exclusively from the slip of screw dislocations gliding on different glide planes in tension and in compression, they can be analyzed quantitatively in terms of the fraction f of plastic strain accommodated by the glide of screw dislocations.^[121] In the case of α -iron single crystals, deformed cyclically at room temperature, f could be determined as $f = 0.03$, 0.06 , and 0.2 for axial strain ranges $\Delta\epsilon_{pl}$ of 5×10^{-4} , 1.5×10^{-3} , and 6×10^{-3} , respectively.^[121] In these cases, the effect of slip plane asymmetry is the major cause of cyclic slip irreversibility. Hence, in the present case, the factor f is in very good approximation identical to the cyclic slip irreversibility p .

In fatigued bcc polycrystals, characteristic distortions of individual grains would tend to occur as a result of the slip plane asymmetry, if the grains were not constrained. These effects are expected to be the more severe, the higher the amplitude, i.e., the larger the contribution of screw dislocations to cyclic deformation would be. On the other hand, in the bulk, these

distortions would not develop easily because of the constraints. However, at the free surface, the shape changes of individual grains can develop more freely. In the case of α -iron polycrystals fatigued at higher amplitudes and at rather high cyclic strain rates, the development of a characteristic surface roughness with the periodicity of the grain size and with stress concentrations, leading to fatigue crack initiation at GBs, has been observed,^[121,131] as shown in Figure 30. The surface roughening and the intercrystalline cracking are clearly apparent in Figures 30(a) and (b). When the fatigue tests are conducted at appreciably lower cyclic strain rates, the deformation behavior becomes more similar to that of fcc metals, as explained previously, and the effect of slip plane asymmetry is reduced, while cyclic strain localization in slip bands and transgranular cracking are enhanced (Figure 30(c)). Magnin and

Driver have obtained broadly similar results on several fatigued high-purity ferritic alloys^[132,133] and have confirmed that, in fatigue tests performed at room temperature, cracking occurs intergranularly, when the cyclic strain rate is rather high, whereas at lower cyclic strain rates, transgranular cracking is dominant. An example from their work is shown in Figure 31 for fatigued Fe-3 pct Si specimens, showing predominantly intercrystalline cracking (Figure 31(a)) at higher and transgranular cracking (Figure 31(b)) at lower strain rates. The overall cyclic slip irreversibility is, of course, expected to vary in the transition from irreversible asymmetric slip at rather high cyclic strain rates in the “low-temperature” regime to irreversible cyclic slip in slip bands at lower cyclic strain rates, at which the glide mode becomes more similar to that in fcc metals. It is, however, difficult to quantify these changes.

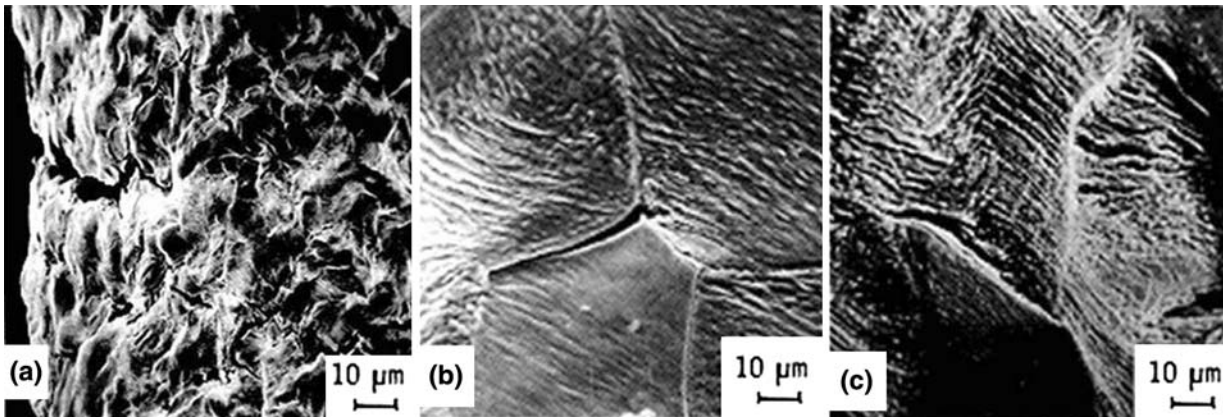


Fig. 30—Dependence of fatigue damage on cyclic plastic strain rate in α -iron polycrystals fatigued until failure at $\Delta\epsilon_{pl} = 1.5 \times 10^{-2}$. (a) Surface roughening and (b) intercrystalline crack initiation at a plastic strain rate $\dot{\epsilon}_{pl} = 4.5 \times 10^{-3} \text{ s}^{-1}$.^[119,121,131] (c) Transcrystalline crack initiation at a plastic strain rate $\dot{\epsilon}_{pl} = 4.5 \times 10^{-5} \text{ s}^{-1}$.^[21,119,131]

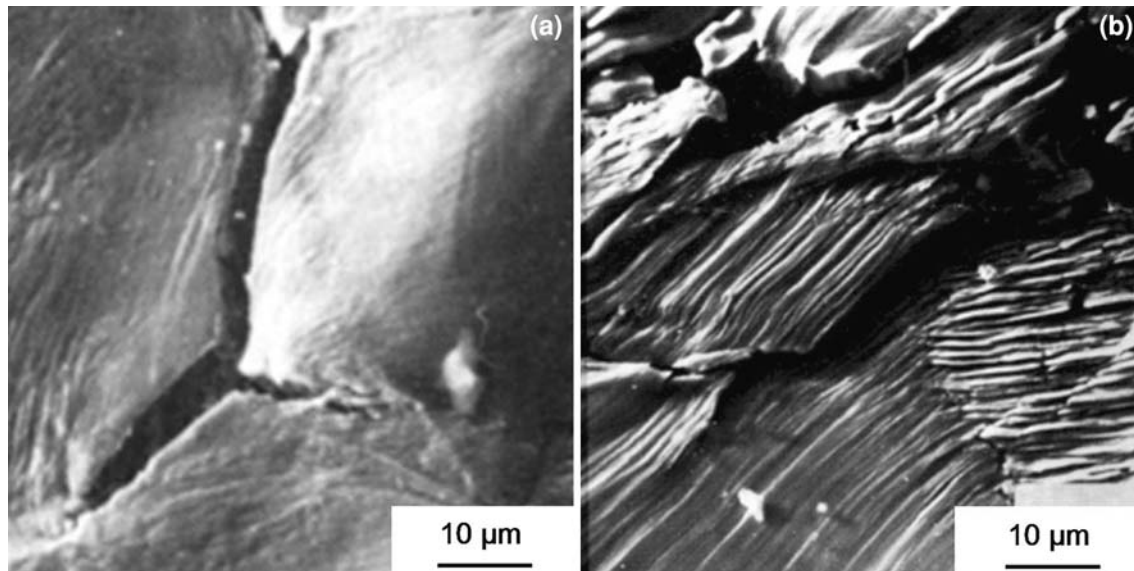


Fig. 31—Dependence of fatigue crack initiation in Fe-3 pct Si fatigued at total strain range $\Delta\epsilon_t = 6 \times 10^{-3}$ on total strain rate $\dot{\epsilon}_t$. (a) $\dot{\epsilon}_t = 2 \times 10^{-3} \text{ s}^{-1}$. Intergranular cracking. (b) $\dot{\epsilon}_t = 2 \times 10^{-5} \text{ s}^{-1}$. Transgranular cracking.^[132] Courtesy of the authors.

Table I. List of Examples of Different Fatigued Materials for Which the Cyclic Slip Irreversibility Could be Estimated*

Example, References	Testing Conditions	Irreversibility p	Evaluation via
PSBs in copper ^[56,88]	$\gamma_{pl,PSB} \approx 10^{-2}$, RT	≈ 0.3	TEM and dislocation model ^[25,26]
UHCF, copper ^[34,35]	$\gamma_{pl,PSB} \approx 1.7 \times 10^{-5}$, RT	≈ 0.000034	SEM and dislocation model ^[34,35]
α -brass X-tals ^[111,112]	$\Delta\sigma/2 = 82$ MPa, RT	≈ 0.005	TEM and computer simulation ^[26]
	$\Delta\sigma/2 = 118$ MPa, RT	≈ 0.03	TEM and computer simulation ^[26]
Cu-2 at. pct Co X-tals ^[114,115]	$\gamma_{pl,PSB} \approx 2.3 \times 10^{-2}$, RT	≈ 1	TEM and dislocation model ^[26]
Waspaloy ^[116]	$\Delta\epsilon_{pl}/2 = 10^{-3}$, 650 °C	≈ 0.05	TEM: irreversible shear displacement ^[26]
α -iron X-tals ^[121]	$\Delta\epsilon_{pl}/2 = 5 \times 10^{-4}$, RT	≈ 0.03	dislocation model of shape changes
	$\Delta\epsilon_{pl}/2 = 1.5 \times 10^{-3}$, RT	≈ 0.06	due to slip plane asymmetry ^[121]
	$\Delta\epsilon_{pl}/2 = 6 \times 10^{-3}$, RT	≈ 0.2	

*The fatigue tests were done either at constant plastic shear strain amplitude γ_{pl} ^[56] or constant axial plastic strain amplitude $\Delta\epsilon_{pl}/2$,^[114,115,121] constant total strain amplitude,^[34,35] or at constant stress amplitude $\Delta\sigma/2$.^[111,112] In order to estimate the cyclic slip irreversibilities, the TEM of bulk dislocation distributions^[25,26,56] was analyzed, or the surface roughness of coated specimens, as observed by SEM^[34,35] or TEM,^[111,112] was evaluated. Then, the cyclic slip irreversibilities could be estimated either in terms of a dislocation model^[25,26,34,35,121] or by computer simulation of random dislocation to-and-fro slip.^[26,34,35] All data except those for Waspaloy refer to fatigue tests at room temperature (RT).

X. SUMMARY OF QUANTITATIVE ESTIMATES OF CYCLIC SLIP IRREVERSIBILITIES p

It has been shown in the preceding sections that quantitative estimates of the cyclic slip irreversibilities p can be obtained under certain assumptions. However, only in a few special cases that were adequately documented experimentally was it possible to obtain numerical estimates of p . These cases, with the relevant test parameters, are summarized in Table I. It is noteworthy that p varies over orders of magnitudes, being almost negligible in the case of very low loading amplitudes, *viz.* very high fatigue lives in the UHCF regime, and very substantial at larger amplitudes, *i.e.*, shorter life times, *e.g.*, in the presence of PSBs in the low-cycle fatigue/HCF regime. It is interesting to note that this wide range of slip irreversibilities is broadly consistent with the slip irreversibilities deduced from surface slip step analysis for a larger number of very different fatigued materials, as listed by Shyam and Milligan.^[42]

XI. CONCLUDING REMARKS

An account has been given of the different kinds of cyclic slip irreversibilities in cyclically deformed metals and how these irreversibilities contribute to the evolution of fatigue damage. In the discussion, different kinds of model and engineering materials, namely, pure fcc and bcc metals and solid solution and precipitation-hardened alloys, with varying glide modes, were considered. The following issues have been considered as particularly important.

1. Cyclic slip irreversibilities, in combination with cyclic strain localization in PSBs, represent the major cause of fatigue damage.
2. Cyclic slip irreversibilities can occur by different mechanisms both at the surface and in the bulk of fatigued metals.
3. Cyclic slip irreversibilities at the surface *and in the bulk* manifest themselves as fatigue damage at the surface.

4. In some well-documented cases, it was possible to quantify the cyclic slip irreversibilities p in the interior and to relate them to surface fatigue damage.
5. The cyclic slip irreversibilities p were found to vary significantly in magnitude from case to case, being almost negligible at low loading amplitudes (long fatigue lives) and close to 1 at high amplitudes (short fatigue lives).
6. In spite of the very small cyclic slip irreversibilities at very low loading amplitudes, the *cumulative* slip irreversibilities attained after very high numbers of cycles can still be very large and ultimately lead to failure.

NOTE-ADDED-IN-PROOF

Only very recently has Professor L.M. Brown (personal communication, 2009) revised his model^[19,84,95] of PSBs. While his arguments differ somewhat from those of the EGM model, his final conclusions lead to a picture that resembles very much the EGM model, shown in Figure 12. And, just like the EGM model, his model predicts both static and continuously growing extrusions but does not predict inclusions.

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